

ME141B Lab Report

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Abstract

The purpose of this lab was to fabricate and characterize microfluidic devices for electroosmotic flow. Three different devices were fabricated and tested, including SiO₂ microchannels, glass microchannels, and PDMS microchannels. Fabrication was completed in the UCSB Teaching Cleanroom using thermal oxidation, photolithography, BHF etching, KOH etching, PDMS molding, and bonding processes. Device characterization was performed using a reflectometer, profilometer, and optical microscope to measure oxide thickness, channel depth, channel width, mold height, and sidewall angles throughout fabrication.

Overall, fabrication was successful and produced microchannels with dimensions close to the design targets. SiO₂ microchannels achieved channel depths near the 4 μm target, while glass microchannels reached depths of approximately 2 μm. PDMS molds produced channel heights near the intended 3 μm depth. However, several samples exhibited deviations attributable to differences in material properties, oxidation rates, and etch rates, as well as possible incomplete oxide removal before KOH etching. Photolithography produced channel widths that closely matched the mask dimensions, and sidewall angles generated by anisotropic KOH etching were found to be close to the theoretical value of 54.7°.

Following fabrication and bonding, current monitoring experiments were performed using platinum electrodes placed in the channel reservoirs. Most devices exhibited only picoamp- to nanoamp-level currents, indicating little to no fluid transport. However, one glass microchannel demonstrated current-monitoring behavior that may be indicative of fluid flow. In addition, one SiO₂ microchannel produced microamp-level currents, suggesting successful electroosmotic transport, although alternative explanations for the observed current cannot be completely ruled out. These results demonstrate that the fabrication methods were capable of producing potentially functional microfluidic devices while also highlighting the importance of process control, bonding reliability, and etch calibration in improving device yield and performance.

1. Introduction

Microfluidics is the study and manipulation of fluids within channels whose characteristic dimensions range from micrometers to nanometers. At these length scales, fluid behavior differs significantly from that observed in conventional macroscale systems. Flow is typically laminar, surface forces become dominant, and interactions between channel walls and the fluid can strongly influence transport. These characteristics have enabled the development of a wide range of technologies in biomedical diagnostics, drug delivery, DNA and protein analysis, thermal management of electronic devices, and lab-on-a-chip systems. As microfluidic technology continues to advance, improvements in fabrication methods, device reliability, and channel design remain critical for expanding the capabilities of these systems.

One mechanism commonly used to drive fluid flow in microchannels is electroosmotic flow (EOF). When an electrolyte solution comes into contact with a charged channel wall, an electrical double layer (EDL) forms near the surface. Application of an electric field along the channel causes ions within the EDL to migrate, dragging the surrounding fluid and generating bulk flow. The magnitude of this flow depends on several factors, including channel geometry, surface properties, electrolyte concentration, applied voltage, and overall device quality. Because microfluidic systems are highly sensitive to surface conditions and fabrication imperfections, careful control of fabrication and bonding processes was essential to obtaining reliable device performance.

The objective of this laboratory was to provide a structured introduction to cleanroom microfabrication, microfluidic device assembly, and experimental validation techniques. Across three laboratory modules, microfluidic devices were fabricated, bonded, and tested to evaluate their ability to support electroosmotic flow. Three different device architectures were produced: SiO₂ microchannels fabricated on silicon substrates, glass microchannels fabricated through buffered hydrofluoric acid etching, and PDMS microchannels casted from silicon molds. Through these fabrication processes, students gained experience with a variety of cleanroom techniques, including thermal oxidation, photolithography, buffered hydrofluoric acid (BHF) etching, anisotropic KOH etching, oxygen plasma treatment, and soft lithography. In addition, several metrology techniques, including reflectometry, profilometry, and optical microscopy, were used throughout the fabrication process to characterize device geometry and verify process success.

An equally important component of the laboratory sequence was device bonding. Although bonding is often viewed as the final fabrication step, it is one of the most critical determinants of device yield and performance. Dust contamination, surface defects, trapped air bubbles, and poor alignment can all lead to leaks, channel blockage, or altered fluid transport behavior. Such defects directly impact the quality of experimental results and can cause significant deviations from theoretical predictions. Consequently, successful bonding requires careful handling and process control to ensure repeatable and reliable device operation.

The laboratory sequence also emphasized the importance of process validation and root-cause analysis. Throughout fabrication and testing, deviations from expected results were investigated through metrology measurements and experimental observations. Variations in oxidation rates,

etch rates, material properties, bonding quality, and channel geometry demonstrated how seemingly minor process changes can significantly affect device fabrication. Likewise, testing highlighted how channel dimensions, surface properties, electrolyte concentration, applied voltage, and bonding integrity influence electroosmotic flow behavior. These experiences provided insight into the challenges associated with achieving high device yield and reproducible results in microfabrication environments.

This report presents the complete fabrication, bonding, and testing process for the three microfluidic device types. Fabrication methods and process results are discussed for thermal oxidation, photolithography, BHF etching, KOH etching, and associated metrology techniques. Bonding approaches and their impact on device performance are then evaluated, followed by current-monitoring experiments used to assess electroosmotic flow. Finally, the report discusses sources of error, process limitations, and potential improvements that could increase device yield and overall microfluidic performance in future iterations.

2. Overview

Three distinct microfluidic device architectures were fabricated, bonded, and tested over the course of this laboratory sequence: SiO₂ microchannels etched into silicon substrates, microchannels etched directly into glass substrates, and PDMS microchannels cast from silicon molds. Each device type followed a different fabrication pathway but shared common process steps, including substrate cleaning, photolithography, wet etching, and PDMS bonding. The complete fabrication sequences for all three device types are summarized in Figures 1, 2, and 3, respectively, and detailed step-by-step procedures are provided in *Appendices A, B, and C*.

For the SiO₂ microchannel devices, fabrication began with silicon substrate cleaving and cleaning, followed by a solvent clean, RCA SC-1 clean, and BHF native oxide strip. A sacrificial SiO₂ layer of approximately 6,000 Å was then grown via wet thermal oxidation in a tube furnace at 1030°C, with oxide thickness verified across all samples using a Filmetrics F20 reflectometer. Photolithography was performed using a Karl Süss MJB3 contact mask aligner with AZ P4110 positive photoresist spun at 4,000 RPM, targeting a 300 µm channel width and 2 cm channel length. A BHF wet etch was then used to remove the SiO₂ in exposed channel regions, with etch time determined from a calibration performed on a sacrificial piece with known oxide thickness. The photoresist was stripped, and anisotropic KOH etching at approximately 80°C was performed to define the silicon microchannels to a target depth of 4 µm. A second BHF dip removed the remaining sacrificial oxide, and a final thermal oxidation step grew approximately 6,500 Å of SiO₂ to form the insulating channel walls. At each stage, feature dimensions, including etch depth, channel width, photoresist thickness, and sidewall angle were measured using a profilometer, and post-oxidation thicknesses were verified with the reflectometer. Optical microscopy was used to assess photoresist development quality and inspect bonded channel integrity. The completed substrates were bonded to flat PDMS slabs, with O₂ plasma activation performed in a descum system to achieve irreversible bonding. Reservoir access holes were created using a biopsy punch, and the bonded devices were annealed on a hot plate at 120°C.

Glass microchannel fabrication followed a shorter sequence. Glass substrates were cleaned and processed through the same photolithography steps as the silicon devices, using the same three-channel mask. Rather than KOH etching, BHF was used to isotropically etch channels to a target depth of 2 µm, with etch time determined from a dedicated glass etch calibration. Channel depth was measured by profilometry after etching, and bonding to PDMS was performed following the same procedure used for the SiO₂ devices.

PDMS microchannel fabrication required first producing a silicon mold. The mold substrates underwent the same cleaning, oxidation, photolithography, and KOH etching steps as the SiO₂ microchannel process, with the exception that a different two-channel mask with 200 µm wide channels and integrated reservoir features was used, and no final thermal oxidation was performed. The target mold depth was 3 µm. After the sacrificial oxide was stripped with BHF, PDMS was mixed in a 10:1 ratio, poured over the mold, degassed under vacuum, and cured at 80°C. The cured PDMS was peeled from the mold, inlet and outlet holes were punched using a biopsy punch, and the PDMS

was bonded to a glass substrate following O₂ plasma activation. Mold geometry was verified throughout using profilometry, including measurements of channel depth, width, and sidewall angle.

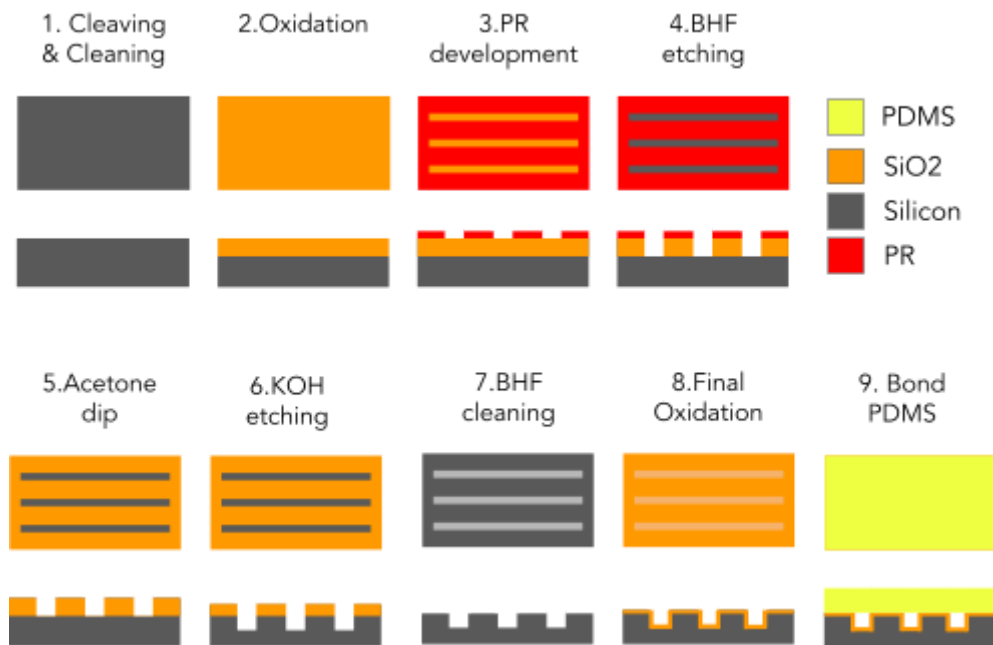


Figure 1: SiO₂ microchannel process diagram. The mask design used to define microchannel geometry with relevant dimensions can be found in the Appendix D.

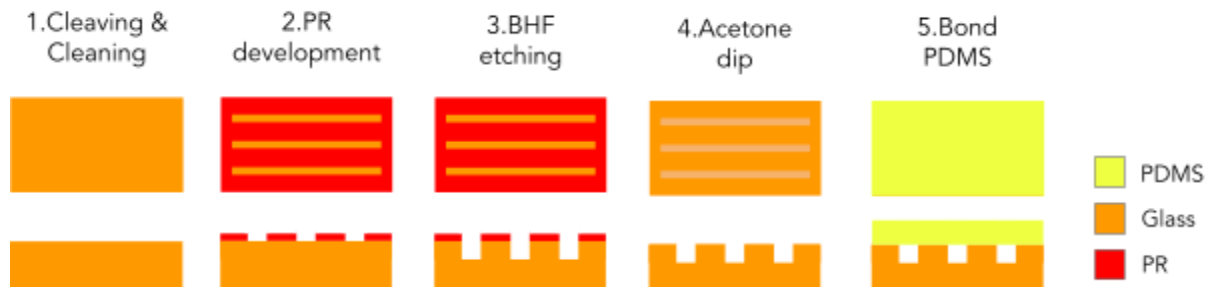


Figure 2: Glass microchannel process diagram. The mask design used to define microchannel geometry with relevant dimensions can be found in the Appendix D.

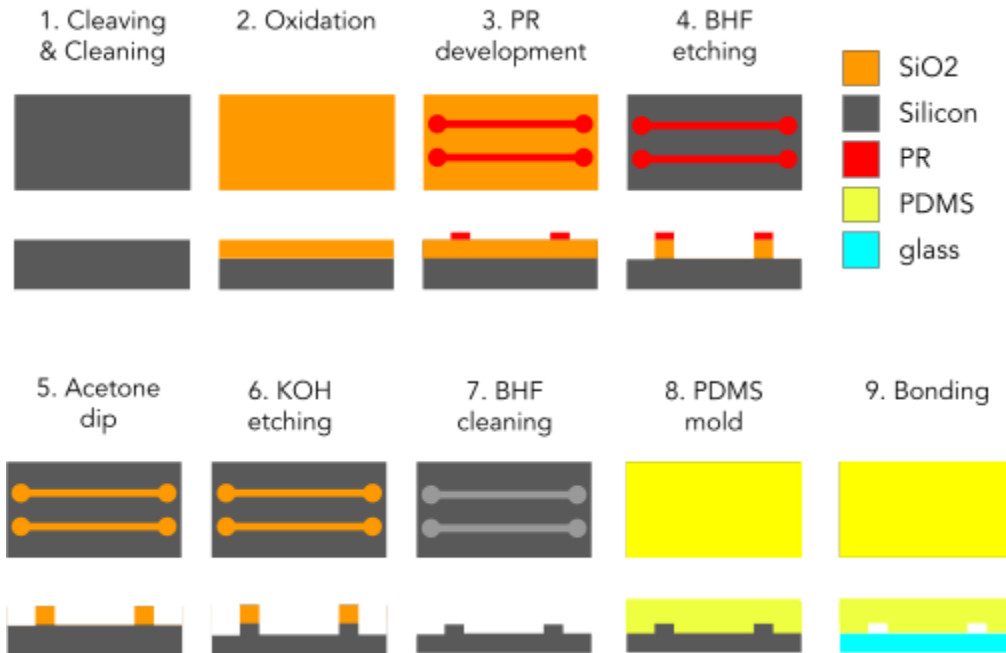


Figure 3: PDMS mold microchannel process diagram The mask design used to define microchannel geometry with relevant dimensions can be found in the Appendix D.

Following fabrication and bonding, all devices were characterized using a current monitoring technique. Two platinum electrodes were inserted into opposing channel reservoirs and connected to a Keithley source-measure unit, with data acquired via a Python script. Applied voltages ranging through 100 V and 500 V were used with electrolyte solution, and current was recorded as a function of time. Full testing procedures are described in Section 6, and detailed fabrication procedures sufficient for reproduction of this work are provided in the Appendices.

4. Process Results

Results for all fabrication processing done for this lab can be seen below. Sections include cleaving, oxidation, photolithography, BHF etching, and KOH etching. This includes all measurements taken, microscope images taken, and any results obtained post-processing to ensure proper fabrication. Measurements were taken using a reflectometer, microscope, and profilometer. Relative profilometer trace areas can be found in *Appendix E*.

4.1 Cleaving

Cleaved samples for all microchannels fabricated can be seen in Figure 4.

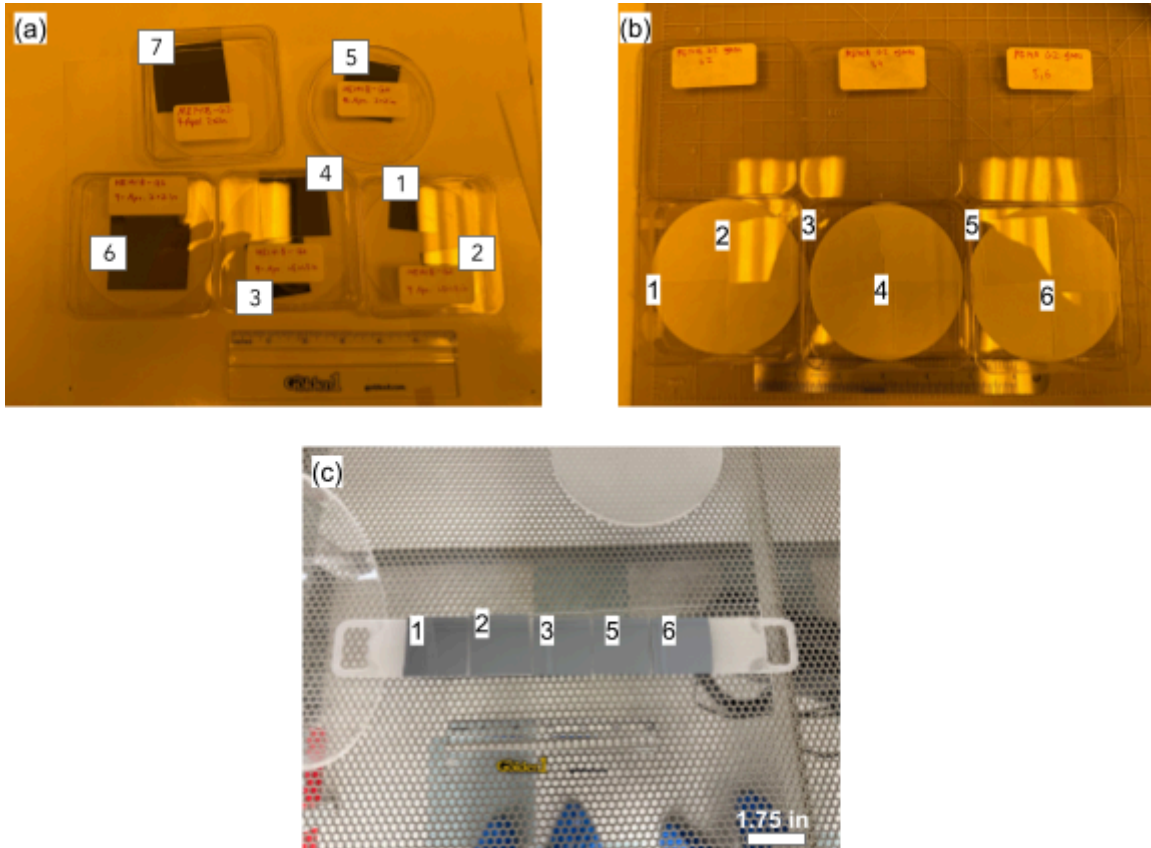


Figure 4: (a) Cleaved silicon substrates for SiO_2 microchannels, labeled 1 through 7, ranging in size from approximately 1.5×1.5 in to 2×2 in. Samples 6 and 7 were provided by the lab manager and consisted of double-sided polished wafers, while the remaining samples were single-sided polished. (b) Six cleaved glass substrates for glass microchannels, labeled 1 through 6, each approximately 1.5×1.5 in. (c) Cleaved silicon substrates for PDMS molds, labeled 1 through 6, excluding substrate 4 due to breakage during the RCA clean. These samples were approximately 1.75×1.75 in. Samples 5 and 6 were fabricated from ultra-thin double-sided polished silicon wafers.

For all substrates, identification numbers were scratched directly onto the back surface of each sample for tracking and identification purposes. A total of five substrates were cleaved for silicon oxide microchannels, six for glass microchannels, and six for silicon PDMS molds.

For the SiO_2 microchannel process, two extra double-sided polished silicon substrates were provided by the lab manager and incorporated. All substrates satisfied the required dimensions of approximately 1.5×1.5 in to 2×2 in.

The glass microchannel substrates were cleaved with an emphasis on maintaining a low aspect ratio to ensure uniform spin coating during fabrication. Each glass sample measured approximately 1.5×1.5 in and was clearly labeled.

Similarly, the silicon substrates intended for PDMS molds were cleaved to dimensions of approximately 1.75×1.75 in while also maintaining a low aspect ratio. Samples 1 through 3 consisted of single-sided polished wafers, whereas samples 5 and 6 were fabricated from ultra-thin double-sided polished wafers. Before oxidation, all silicon mold substrates underwent a solvent clean, RCA clean, and buffered hydrofluoric acid (BHF) clean. During the RCA cleaning process, substrate 4 fractured while being removed from the wafer boat following cleaning and was therefore excluded from further use as a mold substrate.

4.2 Oxidation

Oxidation was performed three times throughout the lab to create microchannels. Oxidation was a required step for the fabrication of SiO_2 microchannels and a PDMS mold. For SiO_2 microchannels, oxidation was used to form a sacrificial oxide layer to complete silicon etching with KOH. After silicon is etched, the remaining oxide is stripped, then placed back into the furnace to create an insulating layer for the microchannel. For the PDMS mold, the oxidation furnace was used solely to form a sacrificial oxide layer to complete Si etching with KOH.

SiO_2 microchannel:

To create SiO_2 microchannels, one of the first steps is to include a sacrificial silicon dioxide layer to create a silicon channel. For the sacrificial silicon dioxide layer, substrates were arranged within the furnace tube in the configuration shown in Figure 5.

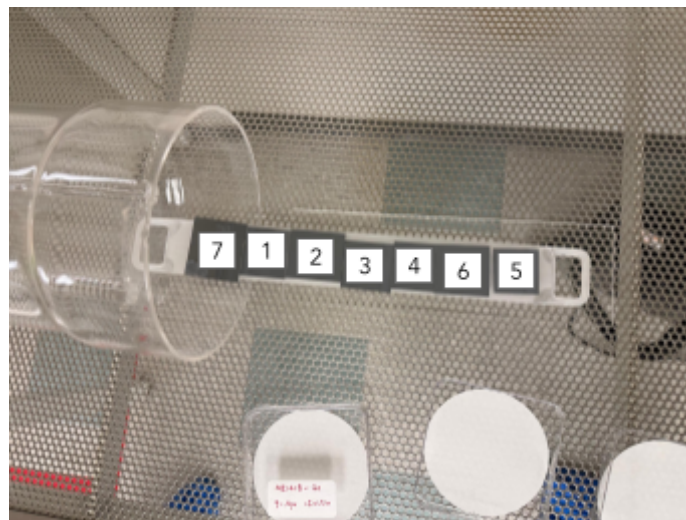


Figure 5: Seven samples positioned inside the furnace before oxidation, labeled according to their location within the tube. Sample 5 was positioned deepest inside the furnace, while Sample 7 was closest to the furnace entrance. The samples underwent wet oxidation for approximately 1.5 hours to produce a target thickness of 6000 Å. This oxidation layer will be used as the sacrificial oxide layer to create SiO₂ microchannels.

Following oxidation, a reflectometer (Filmetrics F20) was used to verify oxidation thickness. For this step, samples were placed into the oxidation furnace for approximately 1:30:40 to achieve a target thickness of 6000 Å of SiO₂, with the required time for oxidation being derived from the BYU oxide growth calculator. The reflectometer was used to measure oxide thickness by directing light onto the sample surface and analyzing the optical interference produced by reflections from the SiO₂ surface and the SiO₂/Si interface. It analyzes the resulting constructive and destructive interference pattern to determine the SiO₂ film thickness and evaluate the quality of fit of the optical model. A schematic illustrating the physics behind the reflectometer is shown in Figure 6.

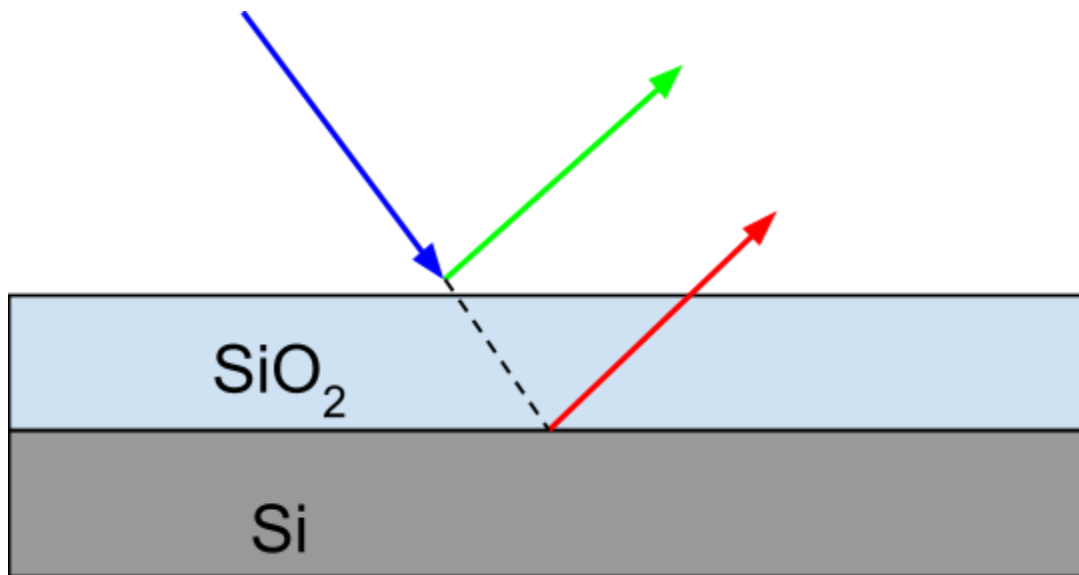


Figure 6: Schematic illustration of the operating principle of reflectometry in a Filmetrics system. Incident light (blue arrow) is partially reflected at the air/SiO₂ interface (green arrow), while the remaining transmitted light (black dashed line) propagates through the SiO₂ layer. This light is then reflected at the SiO₂/Si interface (red arrow) and exits the film stack, where it interferes with the first reflected beam. Interference upon recombination produces the interference spectrum used to determine the film thickness.

Reflectometry measurements were collected from multiple locations on each sample following oxidation. The averaged thickness values are presented in Figure 7.

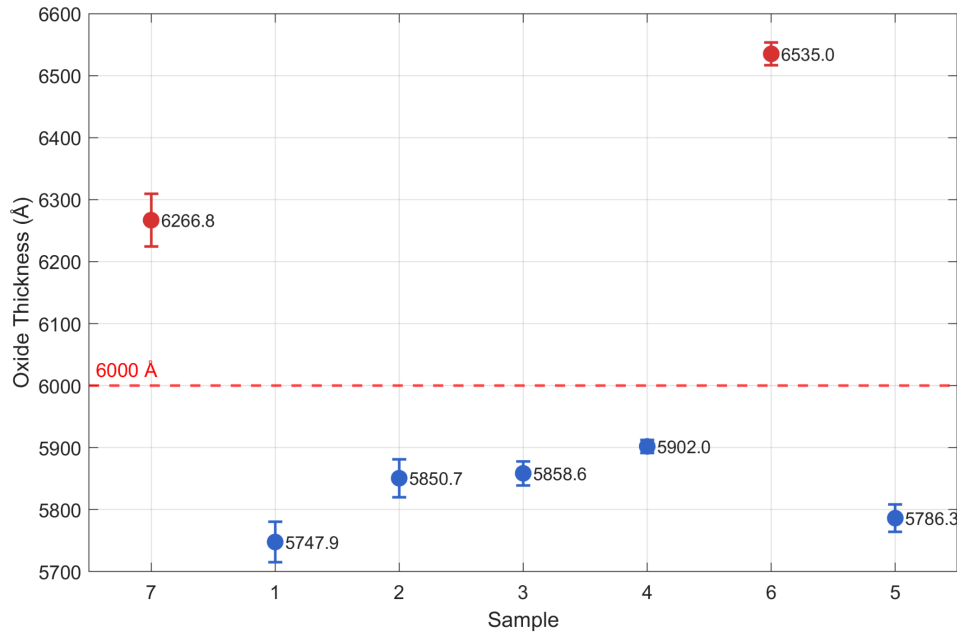


Figure 7: Plot of the averaged thickness from reflectometry measurements of the oxide thickness from each sample. Reflectometry measurements are taken to confirm that the oxidation is successful for all substrates and are used to ensure correct depths are acquired in later steps. Measurements are taken on the top, bottom, center, right, and left regions of each chip. The oxide layer thickness for samples 6 and 7 is noticeably much greater when compared to other samples by approximately 600-800 Å. One thing to note is that samples 6 and 7 are double-polished, provided by the lab manager, and are suspected to be doped, which may explain the higher oxidation rate.

This oxidation was conducted to create a sacrificial layer on silicon oxide microchannels. While the results show thicknesses lower than the target for samples 1, 2, 3, 4, and 5, these thicknesses are still more than sufficient to complete Si etching later in the fabrication process.

As shown in Figure 7, all single-side polished samples exhibited oxide thicknesses slightly below the calculated target value for the specified oxidation time. A trend was also observed across the furnace tube, where oxide thickness increased from Sample 1 toward the center of the furnace, reached a maximum near Sample 4, and then decreased toward Sample 5. This trend suggests improved oxidation conditions near the center of the furnace.

Samples 6 and 7 exhibited higher oxide thicknesses than the remaining samples and did not follow the same spatial trend. Unlike the other substrates, these samples were double-sided polished and were provided separately by the lab manager. Differences in substrate properties, such as doping concentration, may have contributed to the increased oxidation rates observed throughout the fabrication process.

After the silicon microchannels are fully fabricated, the next step is the growth of an insulating SiO₂ layer on top of the bare silicon channel. After an insulating layer of SiO₂ is grown, metrology is used to check channel depth, width, sidewall angle, and oxidation thickness. Metrology results can be seen below in Figure 8.

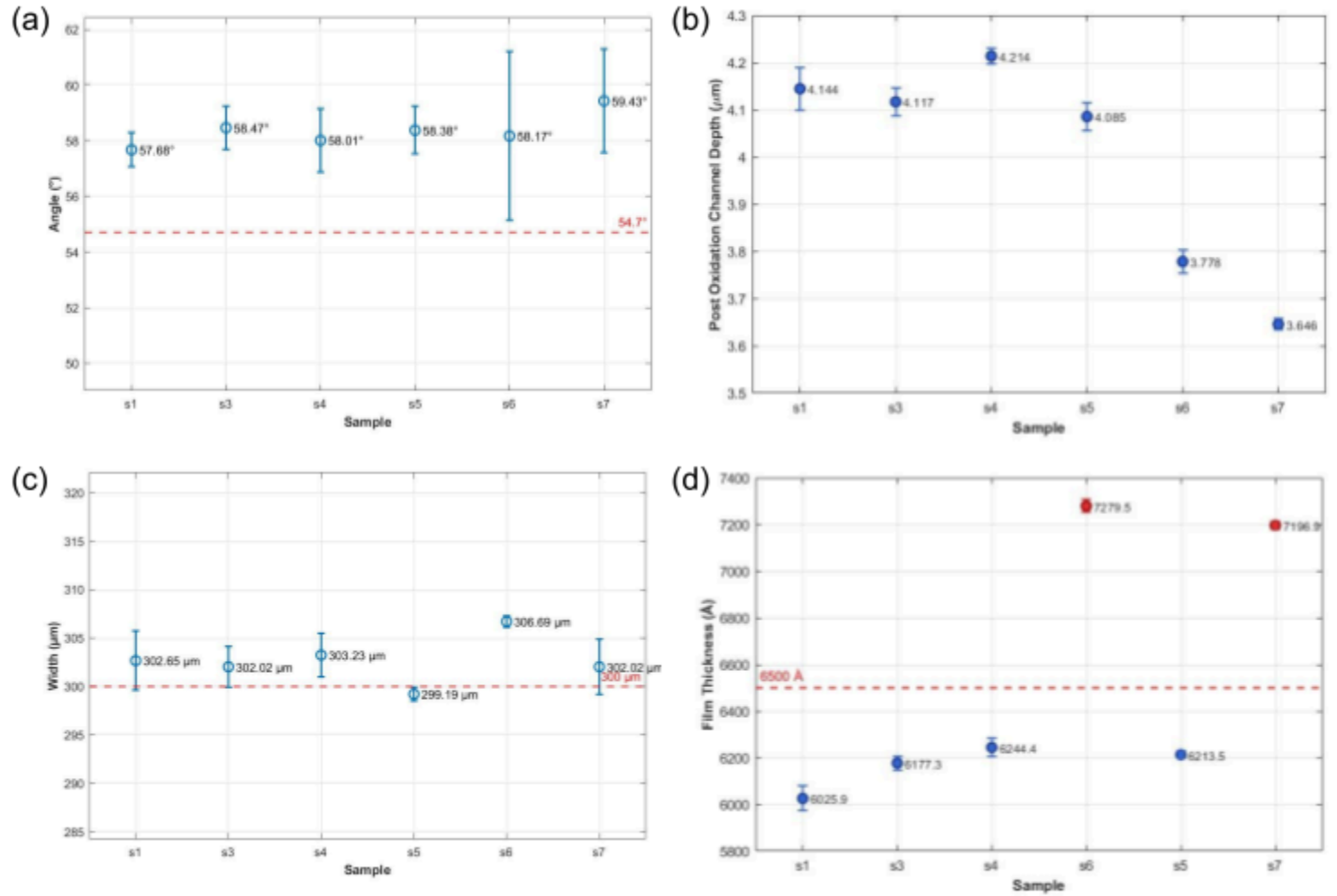


Figure 8: (a) Sidewall angle measurements extracted from profilometry data imported into MATLAB. Angles were determined by identifying the top and bottom points of each sidewall and computing the angle relative to the horizontal. Three profilometry traces per sample (one per channel) were used. Measured angles are consistently approximately 3–4° greater than the theoretical 54.7° expected for anisotropic KOH etching along the $\langle 111 \rangle$ planes. (b) Final channel depth measurements obtained from profilometry, averaged over three microchannel measurements per sample. Samples 6 and 7 show shallower channel depths consistently by about 0.4 micron compared to other samples. (c) Channel width measurements after final oxidation were obtained from profilometry data in MATLAB by identifying left and right channel wall boundaries. The measured widths closely match the designed mask width of 300 μm (red dotted reference line). Differences in measurement methodology compared to earlier width extraction methods may contribute to minor discrepancies. (d) Final oxide thickness measurements obtained using a reflectometer following thermal oxidation. The process consisted of 10 minutes of dry oxidation, approximately 1 hour and 42 minutes of wet oxidation, and a final 10 minutes of dry oxidation, targeting a total oxide thickness of 6500 $\text{\AA}$. Wet oxidation duration was determined using the BYU oxidation calculator, with dry steps assumed to have negligible thickness produced in comparison. Samples 6 and 7 (double-side polished wafers) consistently exhibited approximately 1000 $\text{\AA}$ greater oxide thickness than the other samples.

Across all characterization metrics, systematic trends suggest that substrate properties significantly influenced processing outcomes. The elevated sidewall angles relative to the theoretical 54.7° KOH

etch angle likely arise from combined etch rate variability or measurement uncertainty in extracting wall points from profilometry data.

The reduced channel depths observed in Samples 6 and 7 could potentially be caused by the doped silicon etching behavior, where doping concentration modifies the anisotropic KOH etch rate, resulting in slower etch rate and shallower features under identical processing conditions. Overall, the channel depths are close to the targeted 4 micron depth.

Channel width measurements closely match the designed mask geometry, indicating strong lithographic fidelity. Minor deviations are likely attributable to differences in measurement methodology rather than process variation.

Finally, the increased oxide thickness observed in double-side polished ultra-thin wafers (Samples 6 and 7) is consistent with results from the first oxidation. This furthers the argument that these samples are possibly doped.

PDMS molds:

Similar to the SiO_2 microchannel, a sacrificial layer of oxidation is needed to create the PDMS molds. After oxidation is completed, a reflectometer is used to verify the oxidation thickness achieved. Results can be seen in Figure 9.

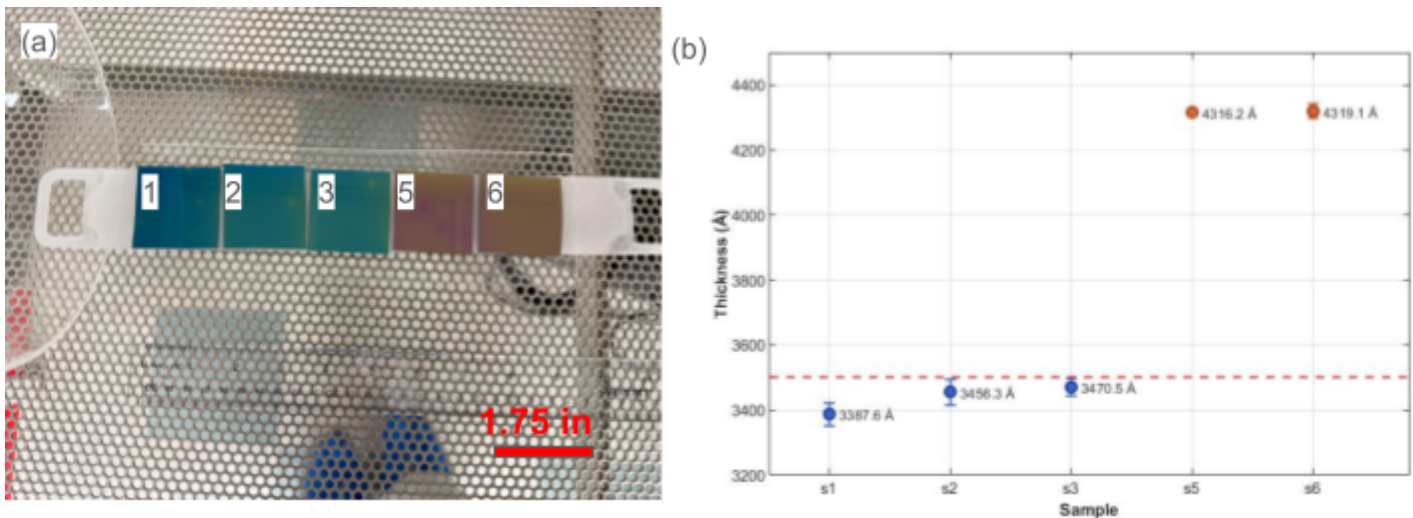


Figure 9: (a) Samples 1,2,3,4, and 6 were subjected to thermal oxidation in a furnace for a wet oxidation duration of 0:38:23, targeting a SiO_2 thickness of 3500 Å. All samples underwent a 10-minute dry oxidation step both before and after the wet oxidation process. Sample 1 was positioned closest to the furnace entrance, while Sample 6 was placed deepest within the furnace, establishing a spatial gradient across the load. Samples 1, 2, and 3 were single-sided polished wafers, whereas Samples 5 and 6 were double-sided polished. Sample 4, also double-sided, polished but fractured during the RCA cleaning process. (b) Post-oxidation measurements with the reflectometer indicate a clear dependence on wafer type. Double-sided polished samples exhibit approximately 800 Å greater oxide thickness than the target value, whereas single-sided polished wafers are significantly closer to the intended 3500 Å growth. This difference suggests that wafer surface condition and possibly doping effects influence oxidation under the given process conditions.

The double-sided polished wafers consistently exhibit a higher oxidation rate, in agreement with trends observed across all oxidation runs in this lab. This behavior may indicate that the wafers are doped, as doping can lead to increased oxide growth rates.

Overall, the single-side polished samples (1–3) produced oxide thicknesses that are closer to the target value compared to other oxidation runs. This improved agreement may be attributed to variations in furnace conditions, as well as differences in wafer loading, such as the number of wafers present in the oven during processing, both of which can possibly influence oxidation rate.

4.3 Photolithography

Photolithography is used for each fabrication process of this lab. In each process, photolithography is used to expose positive photoresist. After development, the sample continues to the next step, which is some type of etch. All masks used during photolithography can be found in *Appendix D*. The difference between the two masks is that the first mask has 3 channels without reservoirs, while the PDMS mold mask has 2 channels with reservoirs. Among those main differences, there are also slight differences in dimensions.

SiO₂ Microchannel:

For the SiO₂ microchannel, photolithography is completed on one sample and evaluated before continuing with other samples. Optical microscopy and profilometry were performed to evaluate the development quality before proceeding with the remaining samples. Sample 1 was selected as the initial verification sample. Microscopy of the developed channels is shown in Figure 10, and photoresist thickness measurements for Samples 1 and 7 are presented in Figure 11.

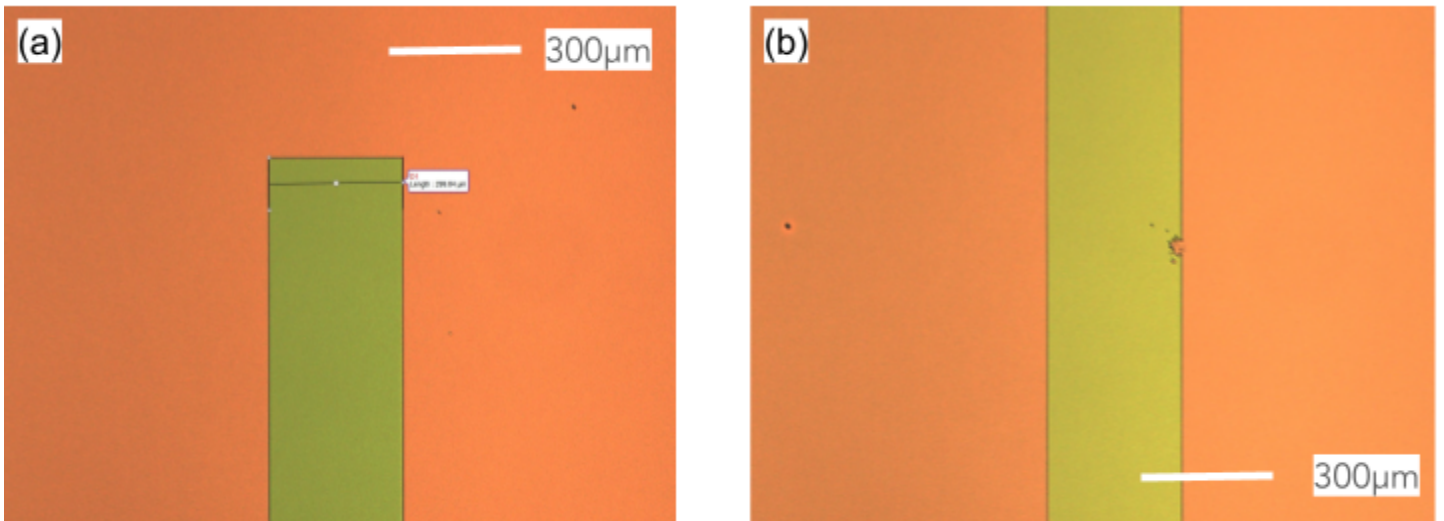


Figure 10: (a) Microscopy on the top of a single channel on sample 1 after photoresist development. The measured channel width is $299 \pm 1 \mu\text{m}$. (b) Microscopy on a portion of the middle channel on sample 1 after photoresist development. The channel has a defect in the middle channel, which consistently appears on all samples.

The mask dimensions seen in *Appendix D* include 300 μm channels, so the microscopy check of photoresist development can be considered successful. While the majority of channels developed as expected, a recurring defect was observed in all middle channels, as shown in Figure 10(b).

After some investigation into the mask, it appeared that the mask has a small bump, which causes this defect in all samples, even after cleaning. This bump is consistent throughout other groups' channels as well.

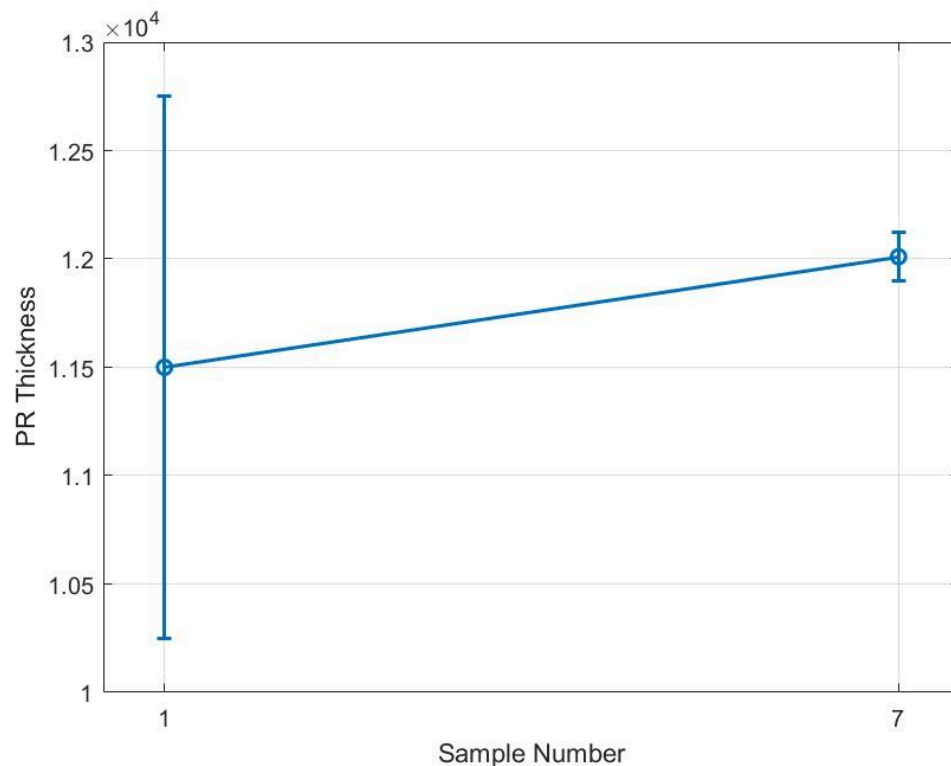


Figure 11: Photoresist thickness measured by profilometry, expressed in angstroms. Sample 1 was evaluated as the primary verification sample. Due to differences in oxide thickness between samples, Sample 7 was subsequently measured for comparison. According to the AZP 4110 photoresist datasheet for the Teaching Cleanroom, spin coating at 4000 RPM for 30 seconds yields a nominal film thickness of 1.0–1.3 μm . Both samples fall within this specification, confirming that development was successful.

After a successful photolithography of sample 1 and sample 7, the leftover samples were developed using the same configurations as sample 1 and sample 7.

Glass Microchannel:

For photolithography of the glass microchannels, the same process for the SiO_2 microchannels was used with the same mask. After spin-coating and development of one glass substrate, the photoresist thickness was measured using a profilometer to determine successful development. After successful development was determined, the rest of the samples were spin-coated, patterned, and developed. Results of measurements can be seen in Figure 12.

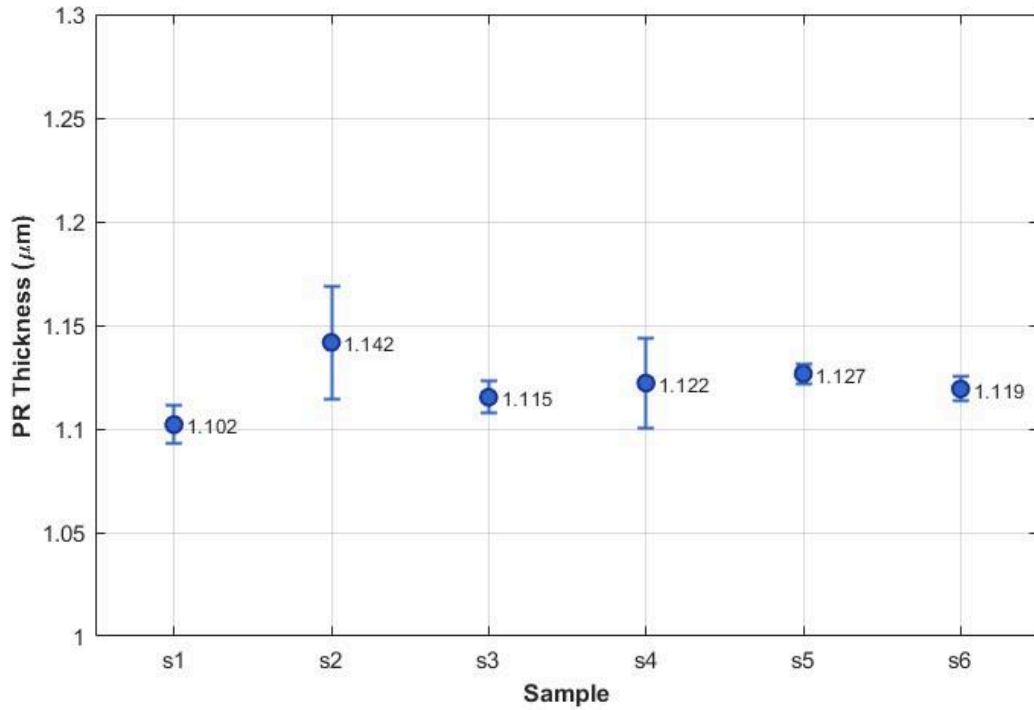


Figure 12: Plot displaying photoresist thickness on glass substrate post-development for samples 1 through 6. PR thickness is about 1.1 μm , which falls within the expected range of thickness for AZP4110 for the given recipe(1-1.3 μm).

Spincoating of photoresist was deemed successful. Additionally, a microscope was also used to ensure that the thickness of the channel matched the mask, which it did, but the image wasn't recorded.

PDMS Mold:

To complete photolithography on the molds, the same process is used, but using a different mask, which can be found in *Appendix D*. Contrary to the other masks that exposed only the channels, this mask exposed everything but the channels to enable the creation of a mold. After the development of one substrate is deemed successful, the photoresist thickness of each substrate was measured using a profilometer to determine successful development across all substrates. Results of measurements can be seen in Figure 13.

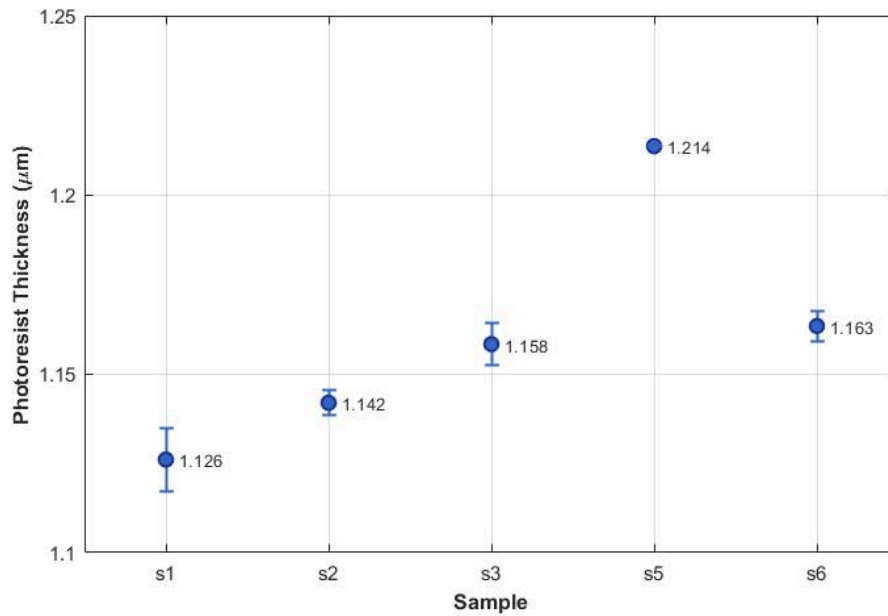


Figure 13: To ensure proper development of the photoresist, a profilometer was used to scan across patterned channels and extract the photoresist thickness. Sample 1 was measured first and used as the initial reference sample before proceeding with measurements on all remaining samples. The measured thickness values for all samples fall within the range of approximately 1.0 to 1.3 μm , which is consistent with the expected film thickness for AZ P4110 photoresist spun at 4000 rpm for 30 seconds.

Spincoating of photoresist was again deemed successful. Additionally, a microscope was used to ensure that the thickness of the channel matched the mask, which was also deemed to match the mask dimensions.

4.4 BHF Etching

During this lab, BHF etching is used as a method to etch SiO_2 isotropically from the samples. BHF etching is done twice for SiO_2 microchannels, once for glass microchannels, and twice for PDMS mold microchannels.

SiO_2 Microchannel:

After developing photoresist, the next step is to complete a BHF dip to remove SiO_2 in regions that don't have the photoresist's protection. Before the BHF dip, a calibration BHF piece was provided by the lab manager. This piece of substrate had $\sim 2000 \text{ \AA}$ of SiO_2 on it and would be used for our calibration. Calibration results are displayed in Figure 14.

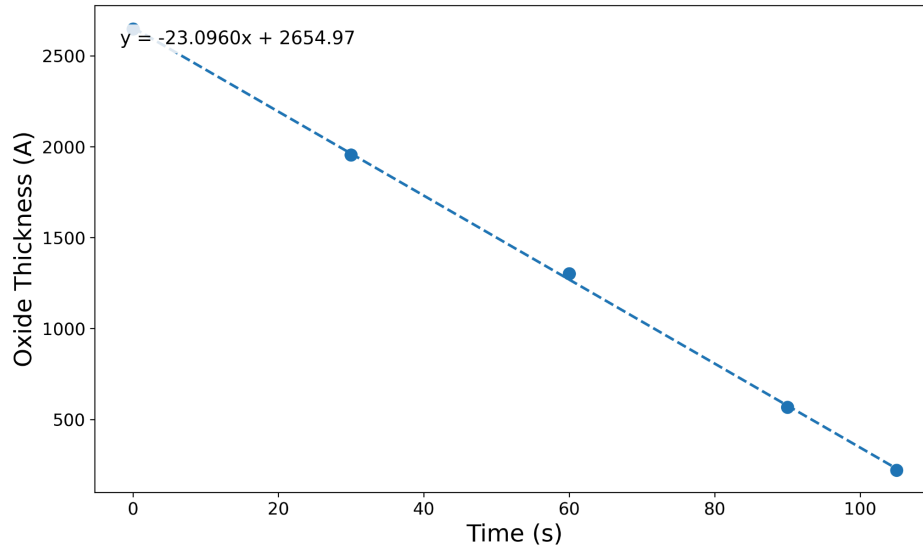


Figure 14: A complementary piece of single-sided polished Silicon wafer with a grown oxide layer was provided as the sacrificial sample to calibrate the BHF rate. Collected data with 4 trials and 3 different sections of the sample measured per trial. Oxide thickness was recorded via reflectometry after a 30 ± 2 second etch for trials 1, 2, and 3. Oxide thickness data from trial 4 were recorded after a 15 ± 1 second etch. Scatter plot of extracted oxide thickness versus etching time, with a dashed line standing for the first-order best-fit line, which has a slope of the etching speed. The slope of the best-fit line indicates the etching speed of BHF, with a unit of Angstroms every second. All BHF etches were conducted with the etching speed derived from the sacrificial sample, with an overshoot.

After etching samples 1 through 5 in BHF for ~5minutes and samples 6 and 7 for ~5.5 min, the total thickness of the PR and SiO₂ layer was measured with a profilometer. Results from measurements can be seen in Figure 15.

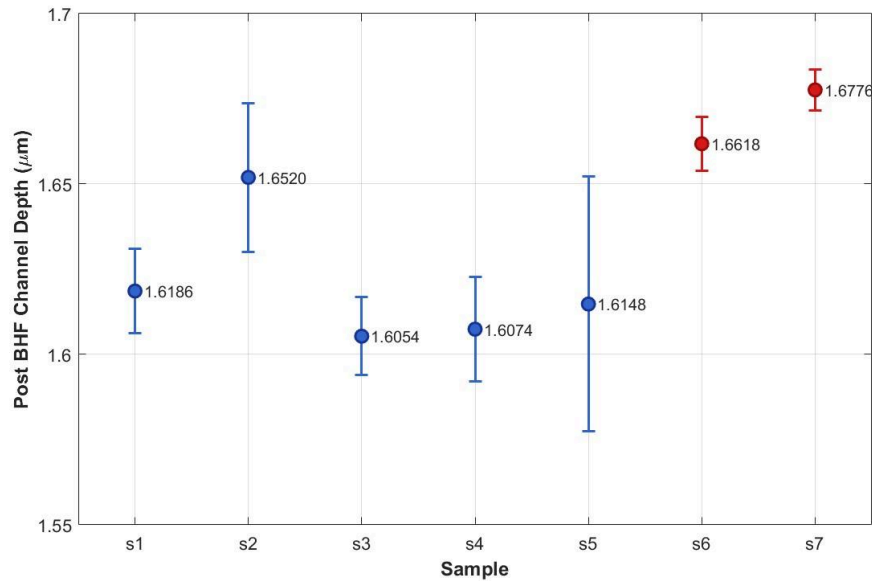


Figure 15: Plot of thickness vs. sample numbered 1 - 7 with error bars. Data was collected with a profilometer and collected through a trial across each channel on the sample. So, a total of 3 channel depth results per sample.

This height is shorter than the estimated PR + Oxide thickness, due to not all the SiO₂ being etched properly, which becomes an issue during the KOH step of the SiO₂ microchannel process. The red dots represent samples 6 and 7, which are double-side polished and contain a larger amount of oxide compared to the other samples. To ensure that a similar amount of oxide was removed from these channels as well, they were dipped for 30 additional seconds.

The height is shorter than the estimated PR and oxide thickness, which should be about 1.7 microns versus the ~ 1.6 microns measured. This means that there could be 0.1 micron of SiO₂ left. When first completing this etch, we were unaware and went onto KOH etching with an SiO₂ layer still left on our channels.

After multiple failed KOH etch attempts with no measurable depth change, we concluded that a residual SiO₂ layer remained in the channel regions, blocking the etchant from reaching the silicon. Since BHF etches SiO₂ isotropically, the absence of photoresist at this stage was not a concern, as the etch could still proceed without it, since the channels were already defined, just not yet etched to depth. With this in mind, all samples were dipped in BHF for approximately 1 minute, targeting just enough time to clear the remaining oxide from the channel floors without excessively thinning the oxide on the surrounding surfaces. After this additional dip, the channels visibly lightened in color, taking on the appearance of bare silicon, which served as a qualitative indicator that the channel floor oxide had been successfully removed. Sample 2 was then used to verify that the BHF correction had worked by performing a test KOH etch, the results of which are presented in the KOH process results section.

After the KOH etch to etch into the Si to create microchannels, a BHF etch is used to remove all leftover SiO₂, so that all that is left is Si. For this step, it is okay to over-dip the samples to ensure all SiO₂ is removed since BHF etches Si negligibly when compared to SiO₂. Due to this, and the substrates having about 4000 angstroms of SiO₂ left, we decided to etch for 5 minutes to ensure all SiO₂ is removed. After all SiO₂ is removed, profilometry and microscopy are used to acquire data on depth, width, and sidewall angle. Measurement results can be seen in Figure 16.

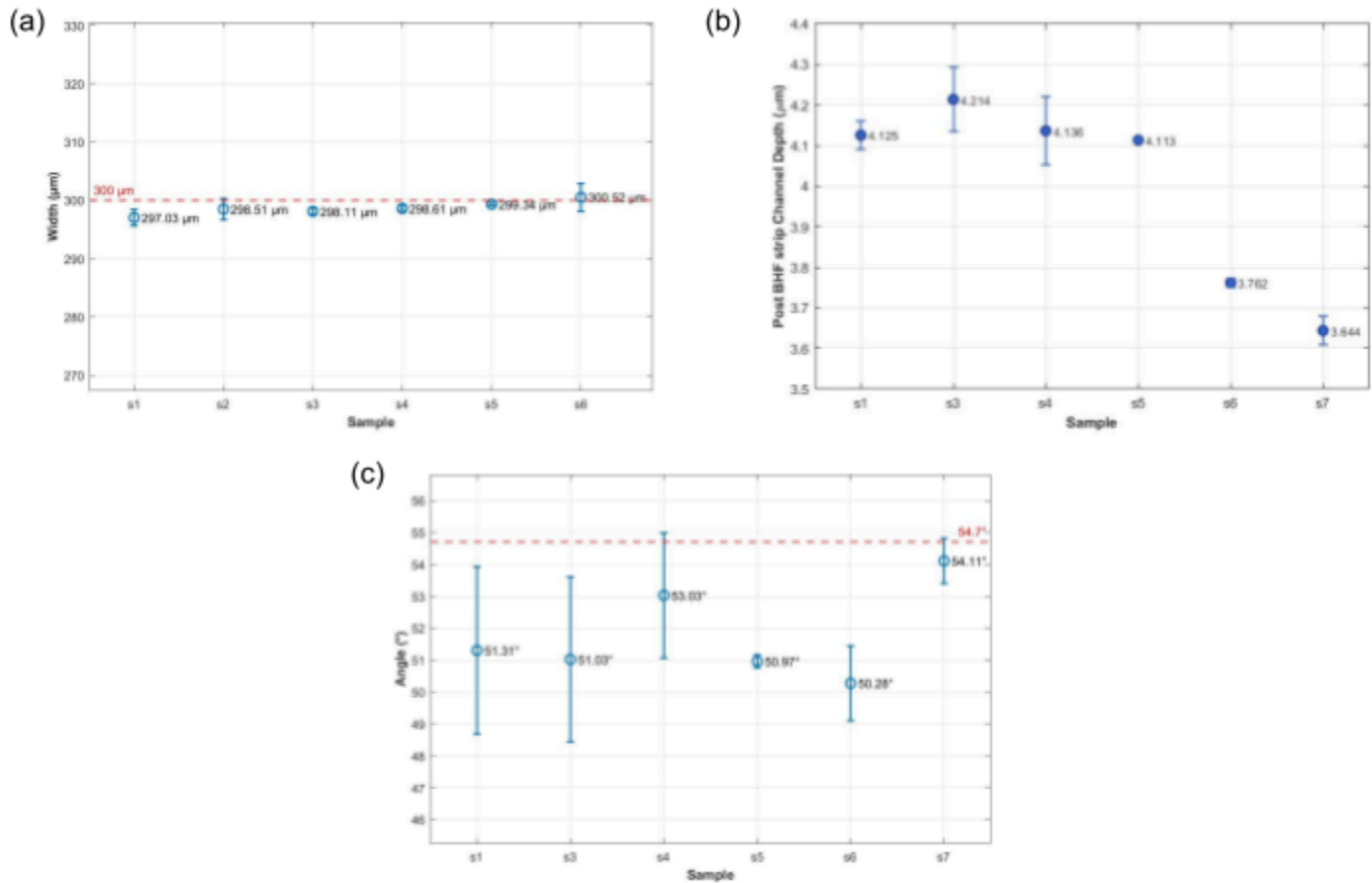


Figure 16: (a) Width measurements after oxide strip measured with a microscope. Width data collected for samples 1,3,4,5,6, and 7 on a microscope. Human error using a microscope can be the cause of any discrepancies. Overall, the data is similar to the width of the channel from the mask, which can be seen as the red dashed line at 300 μm . (b) Plot displaying channel depth data calculated for samples 1,3,4,5,6 and 7 after BHF strip of oxide. Data is in microns and is collected using a profilometer. Samples 6 and 7 are double-sided wafers and potentially doped, which caused the initial oxide thickness to be larger, but could also be the cause of the slower etch rate as well, causing a reduced channel depth. (c) Plot of average sidewall angle for samples 1,3,4,5,6, and 7. Angle was calculated by plotting the profilometer in MATLAB and the user clicking one point at the top of the wall and one at the bottom to calculate the angle from the horizontal. 3 traces from each sample for each channel on the plot were used as profilometer data to estimate the sidewall angle. All average angles are close to the angle predicted from an anisotropic KOH etch (54.7 degrees). The 54.7-degree angle can be seen in the red dotted line.

Overall, the data collected suggests that the channels were successfully fabricated. The width measurements are very close to the mask dimensions of 300 μm , confirming that lithography and etching went smoothly. Along with that, the depth achieved for all channels is close to the targeted depth of 4 μm . Finally, the sidewall angle measurements are very close to the theoretical 54.7°, which just further proves that all etching done with KOH went well.

Glass Microchannel:

To etch channels into glass microchannels, photoresist was developed and used as protection to allow for the etching of microchannels with BHF. To etch channels, a calibration sample was used to ensure that the anticipated depth of 2 μm was reached. The etch calibration can be seen in Figure 17.

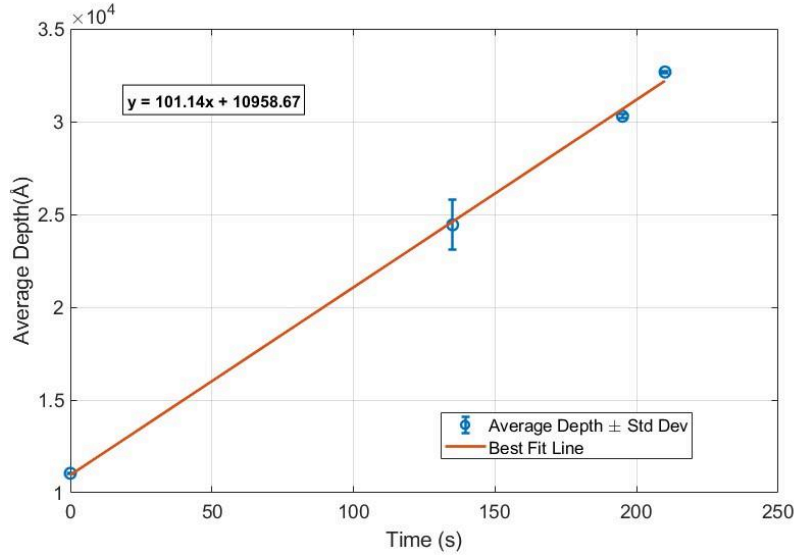


Figure 17: Glass buffered HF (BHF) calibration was initially performed on Sample 1, which was measured before any other samples to establish a baseline etch rate. Sample 1 was then sequentially re-measured after etch intervals of 2 minutes 15 seconds, 1 minute, and 15 seconds. In retrospect, the 15-second dip is likely not reliable, as the transfer time from the BHF solution to the rinse step becomes a significant fraction of the total etch time, introducing substantial timing error and variability.

Following this calibration, Sample 2 was etched to verify the adjusted etch duration. The results indicated that the etch time required an additional ~30 seconds beyond the initial estimate. Sample 3 was then processed and produced satisfactory results, which were used as the reference condition. Consequently, all subsequent samples were etched for 3 minutes 30 seconds, matching the optimized duration determined from Sample 3. Glass microchannel depths can be seen below in Figure 18.

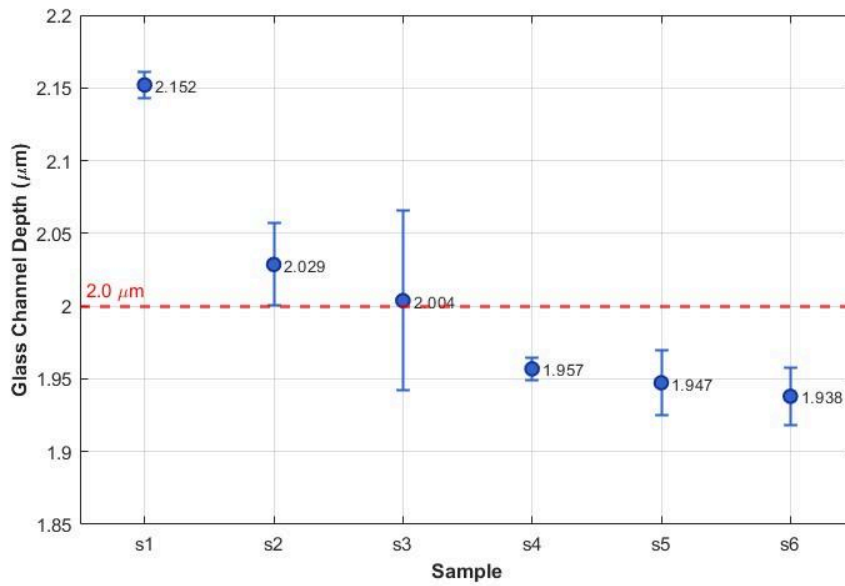


Figure 18: Plot displaying average depth of glass microchannels after BHF dip and solvent clean for all samples. The red dashed line on the plot displays the anticipated target depth for microchannels. Sample 1 was used as the first calibration sample. Sample 2 was used as the second calibration sample, and then sample 3 was used to ensure that the finalized timing after calibration of sample 2 is correct. Samples 4, 5, and 6 were dipped together after the time was finalized with sample 3.

There is a noticeable difference of ~ 0.05 to $0.06 \mu\text{m}$ in depth for samples 4, 5, and 6 when dipped together. One theory for this decrease compared to other samples is that the diffusion rate decreases as more samples are dipped at the same time. Our calibration time didn't account for the diffusion rate to decrease as the sample number increased, which caused samples 4, 5, and 6 to be under the target depth.

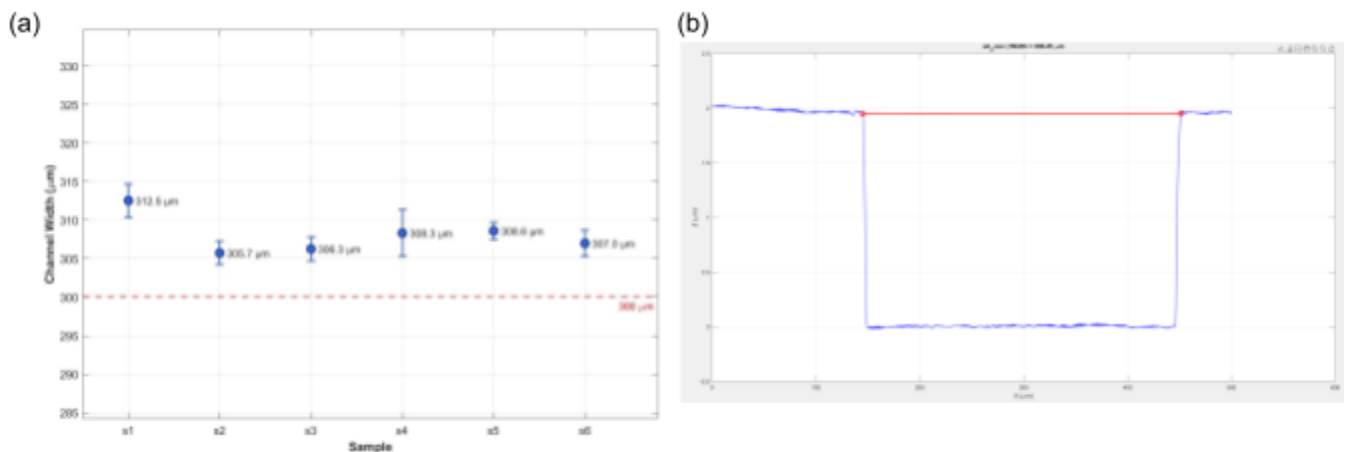


Figure 19: (a) Depicts channel widths for glass samples 1 through 6. Width is calculated using MATLAB, where the user inputs two points, and the horizontal distance between the points is calculated and used to calculate the approximate width. The user clicks the two highest parts of the wall before it flattens. (b) Example of two inputs chosen for MATLAB width measurements for (a).

Results show that the approximate width of the glass channel measured from the top of the walls is slightly greater than 300 microns, which is the anticipated width based on the mask. This can be due to bad contact during exposure, profilometer error during the wall step, and possibly human error using MATLAB to acquire width data from profilometer CSV.

PDMS mold:

Given that this chip is intended to function as a mold, the BHF etch performed to expose the silicon in the regions designated for etching covered a relatively large area. As for this step, over-etching does not pose a concern if dipped for 5-10 minutes, so long as the photoresist doesn't get undercut. So considering this, the dip was carried out for approximately 5 minutes to ensure complete oxide removal before KOH etching. Following this step, profilometry was performed to measure the resulting channel depth and mold height. The corresponding results are presented in Figure 20.

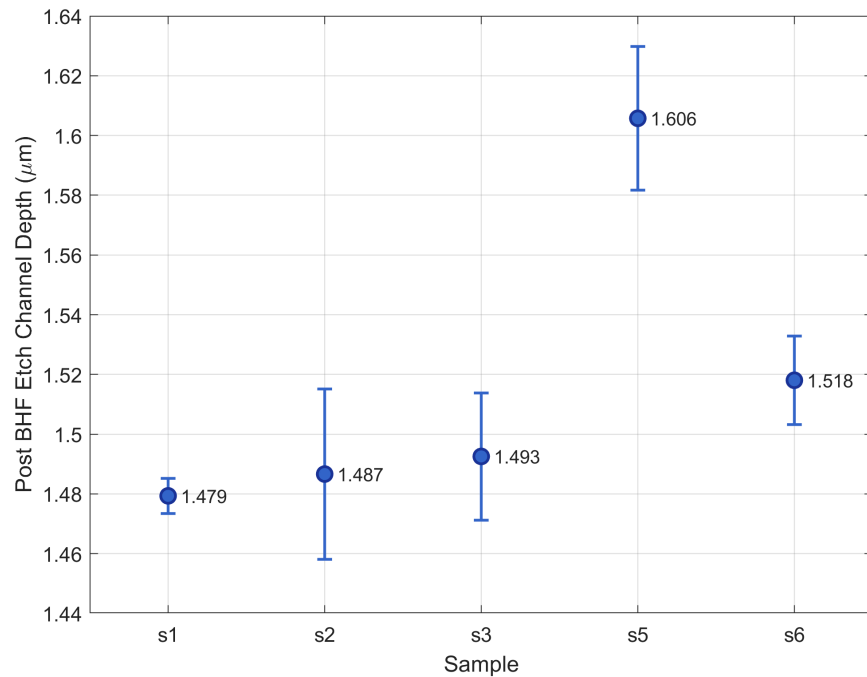


Figure 20: Samples with a sacrificial oxide layer were dipped in BHF for ~5 minutes. Afterwards, the height of the channels was measured by profilometry, and the data presented are the average height of channels A and B on each sample. Samples 5 and 6 have larger heights than the other samples, which is possibly caused by the different oxide thickness during the first round of oxidation due to the material difference. The heights are comparable to the sum of the oxide thickness and the PR thickness, indicating that the oxide uncovered by the PR had been fully stripped off.

Following the KOH etch, all residual sacrificial SiO_2 was removed via a BHF dip lasting approximately 5 minutes. Due to the high selectivity of BHF for SiO_2 over Si, precise timing is not critical for this step.

Once the BHF strip was complete, profilometry was performed to characterize the resulting mold geometry. Measurements of mold height, channel depth, channel width, and sidewall angle were obtained for each sample. The corresponding results are presented in Figures 21 and 22.

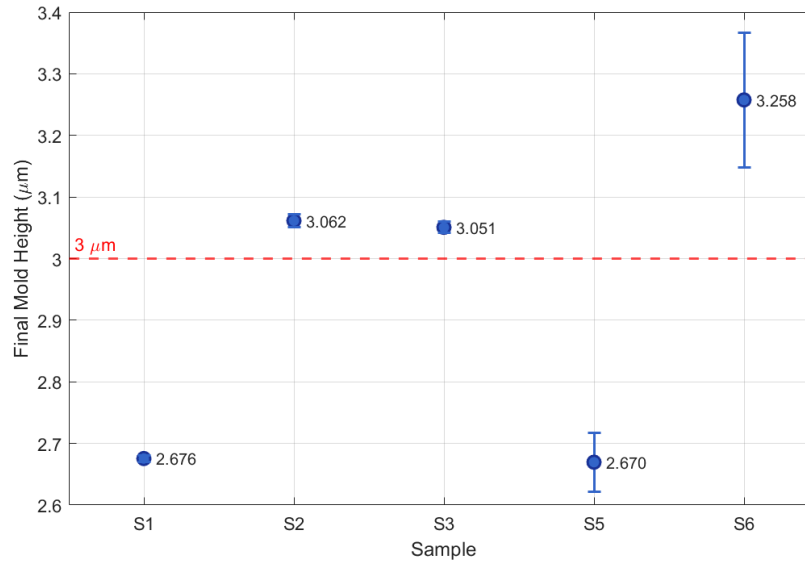


Figure 21: Final channel depth was measured using a profilometer with two scans per sample (one per channel). The target mold height was 3 μm. Samples 2 and 3 are closest to the target and show the most reliable results, while Samples 1 and 5 deviate noticeably. Sample 5 is significantly lower than expected. These variations are likely due to differences in KOH etch rate caused by loading effects (number of samples etched simultaneously) and differences in wafer type and doping. Sample 5, a doped wafer etched alongside Samples 1–3, likely etched at a different rate, contributing to its reduced height. Sample 6 was used as a calibration reference, which may have biased results toward that specific wafer condition.

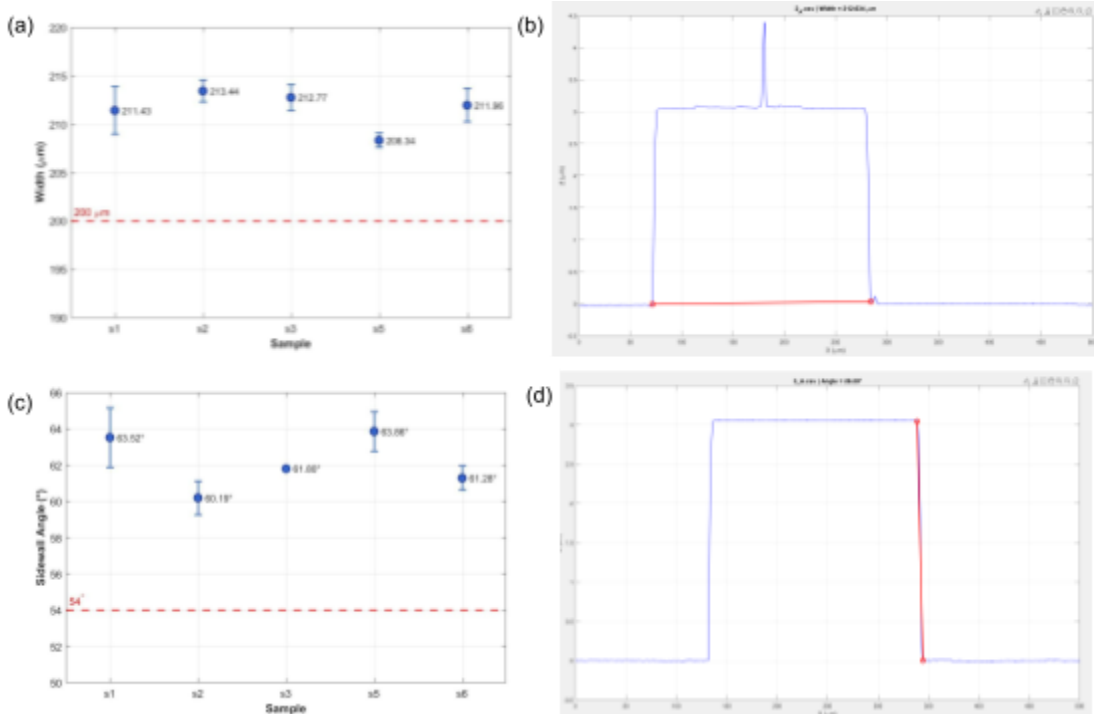


Figure 22: (a) Final channel width measurements on samples 1,2,3,5 and 6. This is done for each channel on all samples. The anticipated width based on mask dimensions is noted by a dashed red line at 200 micrometers. All measurements are greater than the targeted width by about 10 microns. (b) To get width measurements, MATLAB is used to output profilometer data, and the user picks two points, which are subtracted for width. Width errors can be from human errors. (c) Final channel sidewall angle for samples 1,2,3,5, and 6. Red dashed line denotes predicted angle due to KOH etching anisotropically. All sidewall angles are slightly higher than anticipated, which could be from user error for sidewall angle acquisition using MATLAB. (d) Example of how sidewall angle is acquired from profilometer data using MATLAB. Code uses user-picked points to calculate sidewall angle.

Based on the results, the KOH etch was successful for Samples 2 and 3 only. Sample 6, which served as the calibration reference, exceeded the target mold height as a direct consequence of its use in the calibration process. Notably, Sample 6 is double-side polished, whereas Samples 1–5 are single-side polished. This distinction was not accounted for in the calibration, and should be considered in future experiments when selecting a calibration sample.

Samples 1–5 were etched together in a single dip, however, the dip time was not adjusted to compensate for the reduced diffusion rate that results from simultaneously etching multiple samples. This led to a slower effective etch rate than anticipated. These two factors, the mismatch in substrate polish type between the calibration sample and the experimental samples, and the unaccounted diffusion slowdown are the primary sources of the observed deviations from the target etch depth.

Additionally, Sample 1 exhibited a significantly shallower etch depth compared to Samples 2 and 3, despite sharing the same single-side polished substrate. The cause of this discrepancy is not definitively known. However, possible explanations include improper sample placement during the KOH bath, which may have limited solution contact with the channel, or the presence of a residual oxide layer that impeded etch initiation. These possibilities are discussed further in the KOH Etch section.

4.5 KOH Etching

After the sacrificial layer of SiO_2 in the exposed regions of the developed photoresist is removed with a BHF dip, the photoresist is removed with a solvent clean, and then a KOH etch is conducted to etch away at the silicon. One thing to note is that KOH etches anisotropically and creates sidewalls with 54.7° angles from the horizontal. For all microchannels, a calibration was completed with a sacrificial sample before the rest of the samples were dipped in KOH.

SiO_2 Microchannel:

Before KOH etching, a BHF dip was performed to remove all SiO_2 from within the channel, exposing the underlying silicon for subsequent etching. Following the BHF step, the photoresist was stripped in preparation for the KOH dip.

Upon attempting a post-strip profilometry measurement on Sample 2 to confirm etch readiness, no measurable change in height was observed, suggesting incomplete oxide removal. A subsequent KOH dip confirmed the presence of a residual SiO₂ layer, as no silicon etching occurred. In response, the BHF dip was repeated with an intentional over-etch to ensure complete removal down to bare silicon. This was first verified on a single sample before being applied to the remaining samples. While this additional BHF step reduced the thickness of the sacrificial SiO₂ layer, the isotropic nature of BHF etching allowed the process to proceed effectively even in the absence of photoresist. Once bare silicon was confirmed across all samples, Sample 2 was again used as the calibration reference. The resulting calibration data is presented in Figure 23.

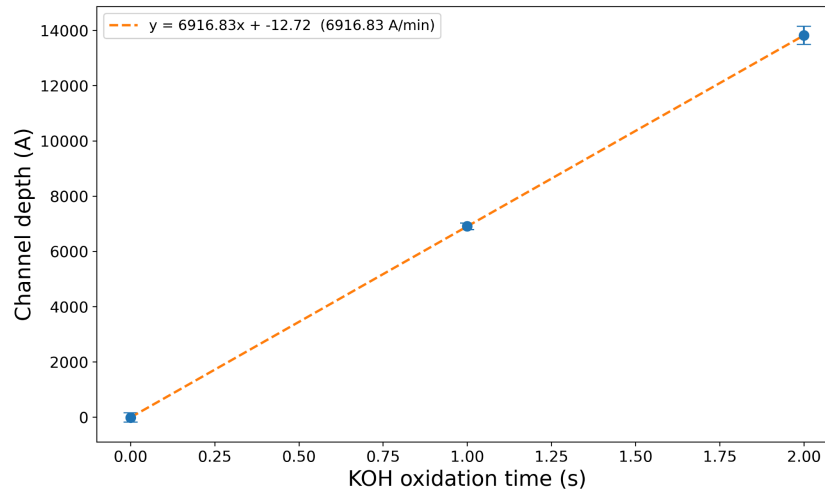


Figure 23: Sample 2 was selected as the sacrificial piece for KOH calibration. It was initially measured via filmetrics and profilometer, and then dipped in KOH for ~1 minute, with both measurements on filmetrics and profilometer repeated for comparison. The change in net drop across the 1-minute dip was recorded and repeated 2 more times. The slope of the plot is extracted as the etching speed of KOH on silicon per minute.

After the KOH etch is completed for all samples, metrology with a profilometer is used to measure channel depth, which can be seen in Figure 24.

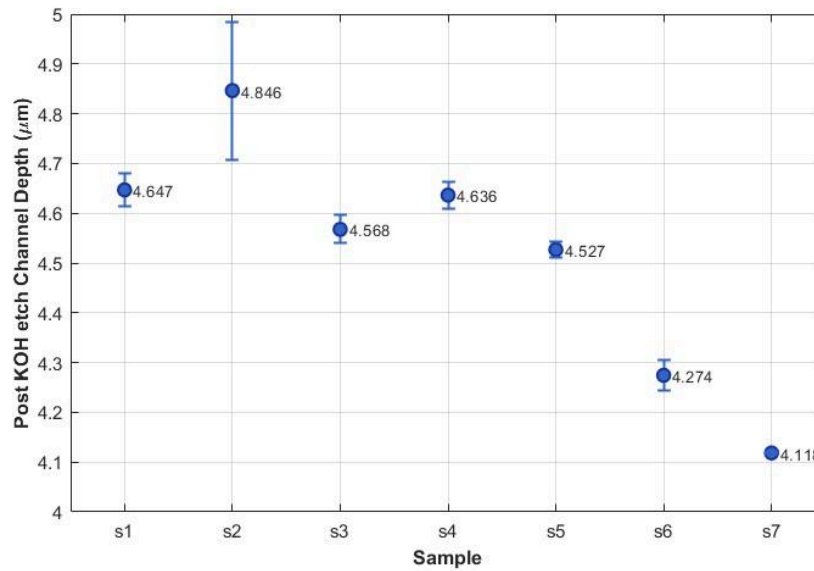


Figure 24: The etching speed from the calibration was utilized, and all other samples were dipped for ~5 minutes and 15 seconds, targeting a channel depth of about 4 μm. Via measurements with a profilometer, all samples of single-side polished (1, 3, 4, and 5), regardless of the sacrificial sample 2, have a channel depth around 4.6 μm, and the double-side polished samples 6 and 7 have a shallower channel depth, around 4.2 ± 0.1 μm. This difference is suspected to originate from the material difference between the samples rather than random uncertainty between samples.

Since the target channel height is approximately 4 μm, the required etch depth for this step must account for both the silicon etch and any remaining SiO₂. Given that the SiO₂ layer grown during oxidation was approximately 6,000 Å (0.60 μm), the combined target height is roughly 4.6 μm, which is roughly consistent with the profilometry measurements obtained for Samples 1 through 5.

Samples 6 and 7, which were double-side polished, yielded different etch depths compared to the single-side polished samples. This discrepancy may be attributable to inherent material property differences between the two substrate types that influence the KOH etch rate.

Overall, the KOH etch was largely successful, particularly given the complications encountered earlier in the process, where premature PR removal occurred prior to complete SiO₂ clearance from the channel interior. The fact that target-range depths were achieved for the majority of samples despite this setback reflects the robustness of the repeated BHF over-etch correction that was implemented.

PDMS mold:

For the PDMS mold, similar to the SiO₂ microchannel, a KOH etch is used to etch away at the silicon. To calibrate the etch rate of the KOH, sample 6 is used as a sacrificial sample. After the KOH etch is completed, the channel depth is recorded with a profilometer, where the results can be seen in

Figure 25.

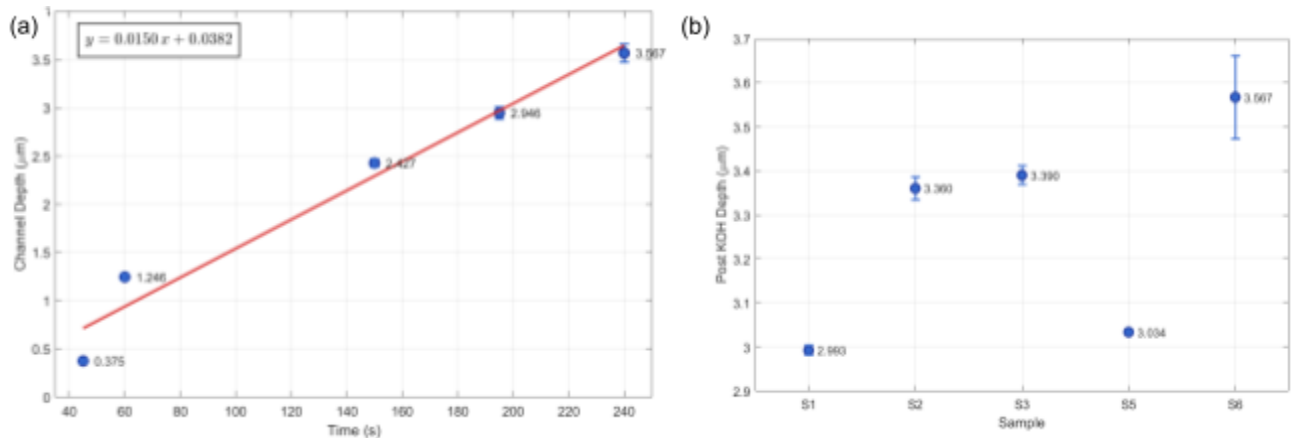


Figure 25: (a) The average channel height measured with the profilometer for sample 6 after each timed KOH etch for calibration purposes. The slope of the best-fit curve is used for the calculation of etching time. (b) All single-side-polished samples were etched for roughly 4 minutes and 10 seconds, and double-side-polished samples were etched for roughly 4 minutes. The final channel heights were also measured via the profilometer. A random uncertainty seems to persist between the trials, which is reasonably suspected to be the diffusion in the KOH bath over prolonged etching periods, which will alter the final thicknesses of channels A and B on each sample, and across samples.

One important consideration for future iterations of this experiment is the implementation of separate calibrations for single-sided and double-sided samples. Additionally, adequate time should be allotted to account for the reduction in diffusion rate that occurs as more samples are introduced into the etch bath. Both of these factors likely contributed to the slower-than-expected etch rate observed for Sample 5, as well as the deviations seen in Samples 2 and 3.

Sample 1 exhibited a notably different etch behavior despite sharing material properties with Samples 2 and 3. Several explanations may account for this discrepancy. First, a thin residual layer of SiO_2 may not have been fully removed during the preceding BHF step, causing the etch initiation to be delayed relative to the other samples. Alternatively, it is possible that during sample loading with the dipper, the channels of certain samples were partially occluded by the backside of an adjacent sample, thereby limiting their exposure to the KOH solution and reducing the overall etch depth. These remain hypotheses pending further investigation.

The KOH etch was performed with a target silicon removal of approximately $3\text{ }\mu\text{m}$. Given that an oxide layer of roughly $0.35\text{ }\mu\text{m}$ remains on the samples, the effective total thickness before the BHF strip is approximately $3.35\text{ }\mu\text{m}$. Samples 2 and 3 fall within this expected range and are considered on target. Based on current measurements, Samples 1 and 5 are anticipated to fall below the target thickness following the BHF oxide strip, while Sample 6 is expected to exceed it.

An image of all mold samples post-BHF etch can be seen in Figure 26.



Figure 26: Image of PDMS molds post KOH etch before being placed into descum. As can be seen, the shiny silver region is the silicon, and the purple and blue regions are SiO₂. After the descum, the molds will all be placed into a BHF bath to remove all SiO₂. This image shows the clear shape that the PDMS molds will hold for microchannels. After SiO₂ is removed, imaging of the channels is difficult to see with a camera as the channel is only about 3 μm deep and the sample will be fully silicon.

5. Bonding

Bonding represents one of the most critical steps in the microfabrication of microfluidic devices. As the final step before a device is considered complete, it directly determines device yield and the quality of data collected through characterization methods such as current monitoring and conductivity testing. A successful bond requires pristine surface conditions, as dust, air gaps, or surface contamination can compromise channel sealing, introduce leakage paths or fundamentally alter fluid behavior. Because the bonding surfaces are fully exposed during this step, even minor handling errors can be detrimental to device performance. Consequently, this process demands the greatest degree of care and patience of any fabrication step.

The bonding interfaces employed in this work included glass-to-PDMS and PDMS-to-SiO₂. These configurations were selected over alternatives such as glass-to-glass or PDMS-to-PDMS primarily due to their relative ease of execution. Each pairing still has challenges, as successful bonding requires uniform contact across the entire interface with no air gaps or voids.

Reversible bonding is achieved without plasma activation by simply pressing the PDMS against the substrate and annealing. This proved to be a reliable and repeatable approach, as it permitted repositioning without permanently committing the bond. PDMS was first cut from its mold (either a flat mold or a microchannel mold) and placed directly onto the target substrate. For SiO₂ microchannel substrates, the PDMS was positioned to cover all channels completely; the same procedure was applied to glass microchannel substrates.

For PDMS microchannel devices, after the patterned PDMS was cut from its mold, inlet and outlet access holes were created using a biopsy punch of appropriate diameter. Punching was performed as carefully and expeditiously as possible to minimize the time during which the PDMS contact surface was exposed to airborne particulates. Following hole creation, the PDMS was carefully placed onto a solvent-cleaned and dried glass substrate.

For SiO₂ and glass microchannel substrates, once the PDMS was positioned, a biopsy punch was used to define reservoir locations aligned to the channels. This step was difficult as the punching of the PDMS was directly on the SiO₂ or glass substrate, so that means when using the biopsy punch, lots of care went into ensuring that too much pressure wasn't applied to cause the substrate to break.

Once all channels were covered, a pair of tweezers was used to apply gentle pressure to the PDMS in regions away from the channels, encouraging conformal contact and expelling any residual air gaps, while taking care to avoid applying force directly over channel features to prevent collapse. Following visual inspection to confirm the absence of air bubbles, the bonded substrates were transferred to a hot plate set to 120°C and annealed for five minutes to strengthen adhesion. Substrates were then removed and allowed to cool before being returned to their corresponding petri dishes for storage.

Irreversible bonding was also performed on a subset of substrates. The procedure followed the same steps as described above, with one critical addition: immediately after peeling the PDMS from its mold, both the PDMS and the target substrate were individually subjected to O_2 plasma treatment in a descum system for two minutes. This exposure activates the surfaces by generating silanol ($Si-OH$) groups, which form covalent $Si-O-Si$ bonds upon contact, producing a permanent seal. For PDMS microchannel devices, biopsy punching was completed prior to plasma treatment to avoid re-exposing the activated surfaces to contamination.

Bonded substrates for all types can be seen in Figure X.

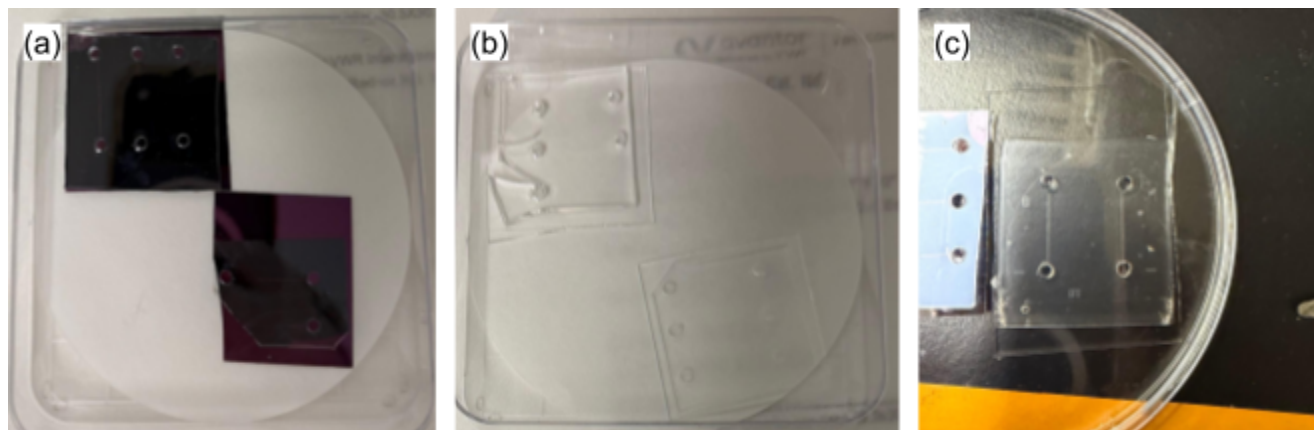


Figure 27: (a) PDMS bonded to SiO_2 microchannels. Sample 3 is the top-left sample and sample 4 is on the bottom right. 3 channels were created on sample 3, but only one channel was created on sample 4. (b) Glass samples 5 and 6. Sample 5 in top left and sample 6 in bottom right. Sample 6 has 2 channels with reservoirs. Biopsy punch resulted in ripped PDMS. (c) PDMS mold microchannels on glass substrate. The sample 3 mold is used to create this channel.

Several failures were also observed. Hole-punching using the biopsy punch frequently caused tearing or distortion of the PDMS, and in some instances, the punch did not produce a clean cut, resulting in partial channel occlusion that prevented fluid flow. There were also cases in which the punch missed the channel entirely, necessitating re-bonding.

In addition, microscope imaging suggested that debris from the punching process may have been introduced into the channel in some samples. Example debris can be seen in Figure 28.

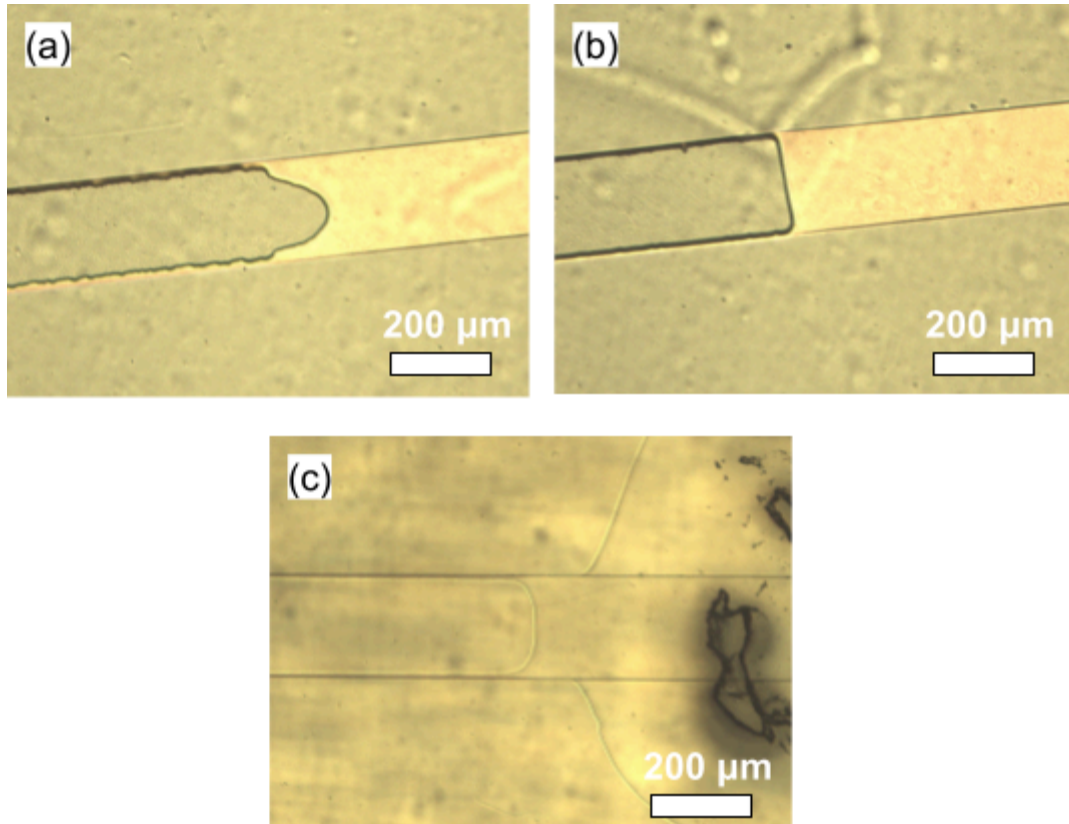


Figure 28: (a) Microscope image displaying collapsed PDMS in SiO₂ sample 3 middle channel. (b) Microscope image displaying collapsed PDMS in SiO₂ sample 3 top channel. (c) Microscope image displaying what is believed to be air gaps and poor bonding on Glass microchannel sample 6 on top channel.

Additionally, while plasma activation was performed prior to bonding for certain samples, it is unclear whether sufficient surface activation was achieved to consistently produce irreversible bonds. Several samples intended to be irreversibly bonded appeared susceptible to peeling, suggesting that bond strength may have been compromised. One possible cause is dust or surface contamination interfering with the bonding process. After observing these issues, subsequent devices were fabricated using reversible bonding methods, which provided more consistent and reliable results under the available fabrication conditions.

Visual inspection of bonded devices revealed several qualitative indicators of bond quality. Well-bonded regions displayed clear channel edges with no visible contamination. Poorly bonded regions exhibited air gaps visible as iridescent interference patterns similar to thin-film optical effects, indicating incomplete contact between surfaces. In some channels, fluid was introduced by placing a water droplet and pushing it in with a thumb. Only for some of the PDMS channels created were we able to see the water flow in a parabolic shape under the microscope. An image of the water parabola can be seen in Figure 29.

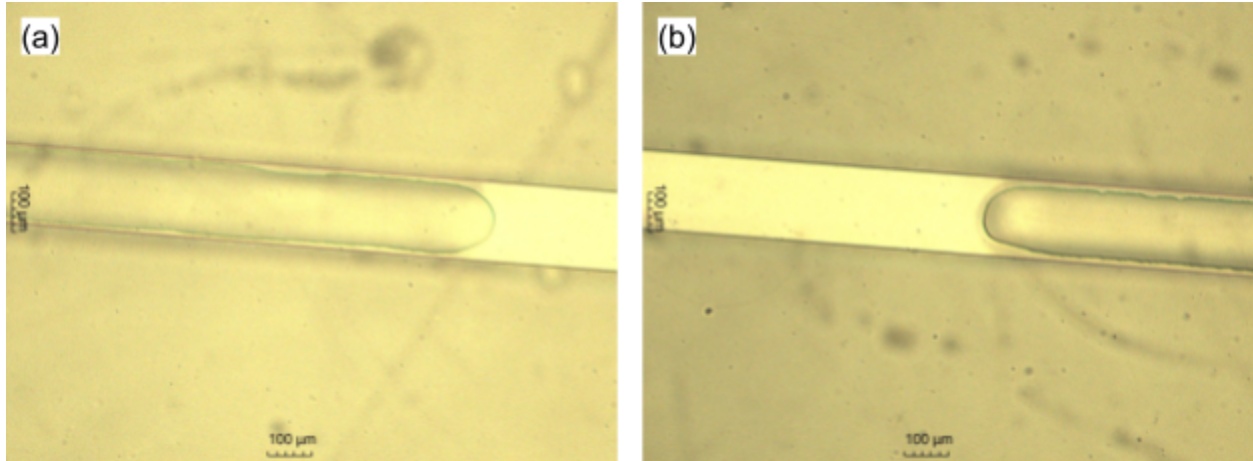


Figure 29: (a) Microscope images of Water parabola in PDMS microchannel sample 3 Channel A. Water droplet is coming from the left reservoir and traveling to the right. (b) Same sample and channel microscope image with water, but coming from the right reservoir. This gives hope that this channel can be used for current monitoring testing and conductivity tests.

Looking ahead, several improvements could increase bond reliability and reproducibility. From a procedural standpoint, including a uniformly flat top surface during curing on a hotplate, would improve the uniformity of PDMS coverage and facilitate cleaner cutouts. Practicing with the biopsy punch before using it on actual substrates could eliminate the tearing and debris introduction observed in this work. Incorporating reservoir features directly into the mold design, rather than relying on post-bonding punched holes, would simplify the process and reduce the risk of channel damage. Finally, more rigorous plasma activation protocols, including higher power and longer descum time, combined with immediate bonding after activation, would increase the likelihood of achieving irreversible bonds where required.

6. Testing

Current monitoring was used to characterize fluid transport through the fabricated SiO₂, glass, and PDMS microchannels. Table 1 summarizes the theoretically expected currents across all channel types and KCl concentrations used throughout testing, calculated from approximate channel geometry and bulk KCl conductivity at 100 V and L = 2 cm. These values serve as the reference against which all measured currents are compared. As shown, expected currents range from approximately 0.039 μ A for PDMS channels at 1 mM KCl up to 2.01 μ A for SiO₂ channels at 22 mM KCl, with the glass channel falling between the two at each condition due to its intermediate cross-sectional geometry.

Concentration (mM)	Conductivity (μ S/cm)	SiO ₂ Current (μ A)	Glass Current (μ A)	PDMS Current (μ A)
1.0	147.0	0.0914	0.0475	0.0393
1.1	161.7	0.1005	0.0522	0.0432
10.0	1470.0	0.9137	0.4745	0.3933
11.0	1617.0	1.0051	0.5219	0.4326
20.0	2940.0	1.8274	0.9490	0.7866
22.0	3234.0	2.0102	1.0439	0.8653

Table 1: Calculated KCl conductivity and expected microchannel currents at V = 100 V, L = 2 cm, across channel types and KCl concentrations.

Before testing, all samples were submerged in DI water and left overnight to ensure full hydration of the channel walls. At the start of each test, fluid was evacuated from both reservoirs using a micropipette, after which both reservoirs were filled with 10 mM KCl solution.

The current monitoring setup consisted of two platinum electrodes inserted into opposing reservoirs, connected to a Keithley source-measure unit that applied a fixed voltage and recorded the resulting current. Data acquisition was controlled via a Python-based script that accepted user inputs for applied voltage, acquisition duration, and file name, and logged the current as a function of time. All initial measurements were conducted at 100 V with a two-minute acquisition window. The full current monitoring set up used can be seen below in Figure 30.



Figure 30: Current monitoring setup located in the Pennathur Lab in ESB. A computer is used to run a Python script that controls data acquisition and measurement procedures. The two red alligator clips connected to the white box are attached to platinum wires, which are positioned within the device electrodes during testing. These wires are connected to the Keithley source meter, which applies the desired voltage and records current measurements. Additionally, the micropipettes shown in the image are used to dispense electrolyte solutions into the sample reservoirs prior to testing.

Electrode placement into the reservoirs proved to be one of the more challenging aspects of the experimental setup due to the small reservoir dimensions; the platinum electrodes are very thin and difficult to position precisely. Along with that, managing the alligator clip connections added further difficulty. A 20 μL micropipette was used to dispense KCl solution into each reservoir by contacting the pipette tip to the reservoir floor before dispensing, ensuring complete filling without air entrapment. Care was taken throughout to maintain consistent electrode contact during solution introduction.

Initial measurements across all channel types consistently yielded currents in the pico- to nanoampere range, indicating little to no ionic transport through the channels. All samples were returned to DI water and retested the following day, with no improvement observed. Representative current monitoring traces are shown in Figure 31.



Figure 31: Current monitoring results under initial 100 V conditions. (a) DI water-filled PDMS channel. (b) 10 mM KCl glass microchannel.

To initiate electroosmotic flow, the applied voltage was increased to 500 V. At this elevated field, bubble formation was observed in one reservoir of the SiO_2 microchannel sample, and the measured current increased to the microampere range, indicating that fluid transport had possibly been initiated. These results are shown in Figure 32.

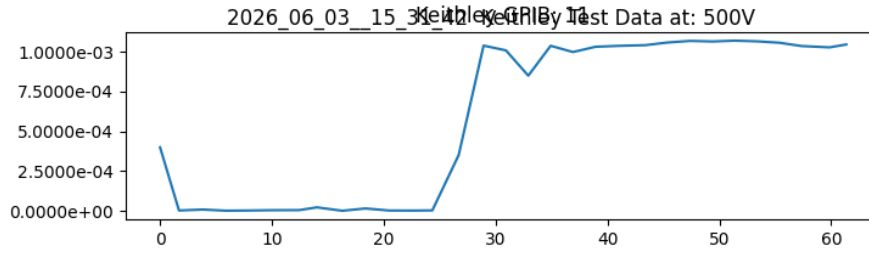


Figure 32: Current monitoring trace at 500 V showing microampere-level current, indicative of electroosmotic flow in the SiO_2 microchannel.

After applying 500 V, the measured current stabilized at approximately 1 mA, which is several orders of magnitude greater than the expected value, despite the applied voltage being only five times higher than the intended operating voltage. This discrepancy suggests the presence of an unintended current path, likely caused by electrical leakage through the device or surrounding materials rather than solely through the electrolyte-filled channel.

To further investigate device behavior, the applied voltage was reduced to 100 V. At this lower voltage, microampere-level currents were still observed. A representative current trace showing a stable signal of approximately $0.95 \mu\text{A}$ over a one-minute acquisition period is presented in Figure X. Based on the channel geometry and the conductivity of 10 mM KCl ($1470 \mu\text{S}/\text{cm}$), the theoretically expected current at 100 V is $0.914 \mu\text{A}$, which closely matches the measured value.

Although this agreement between theoretical and experimental current initially suggests that the channel may be functioning as intended, the shape of the current trace indicates otherwise. Specifically, the observed decay in current over time is inconsistent with the steady-state behavior expected for purely conductive flow through the electrolyte. This behavior suggests that leakage currents or alternative conduction pathways may be contributing to the measured signal. Therefore, while the magnitude of the current agrees with theoretical predictions, the transient response indicates that the measured current may not be exclusively flowing through the fluid-filled microchannel.

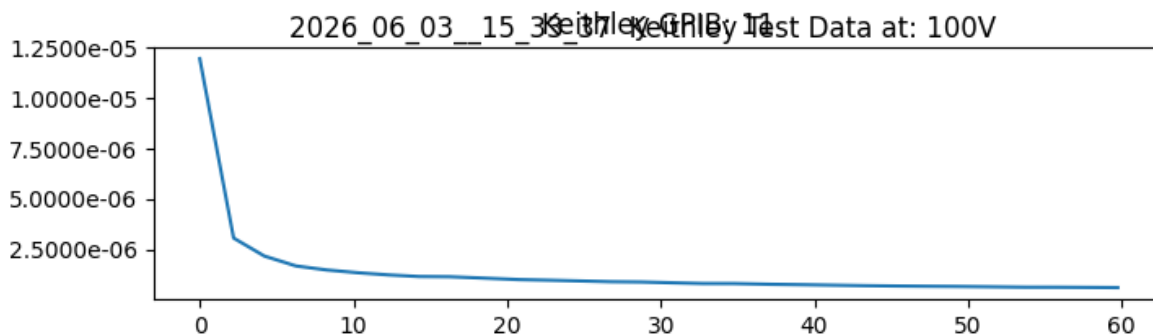


Figure 33: Current monitoring at 100 V following high-voltage initiation, showing a stable $\sim 0.95 \mu\text{A}$ response compared to a theoretical prediction of $0.914 \mu\text{A}$.

This response was not replicated in any other channel type or sample. All remaining SiO_2 , glass, and PDMS channels continued to yield only pico- to nanoampere currents under identical conditions. Notably, no bubble formation was observed at the electrodes during any of these measurements, in contrast to the behavior seen in the SiO_2 channel. It should also be noted that the current trace from the SiO_2 channel did not exhibit the waveform expected of successful electroosmotic flow, which typically rises from a baseline and stabilizes at a steady plateau as the ionic concentration front traverses the channel. Instead, the trace displayed an exponential decay profile, more consistent with capacitive discharge at the electrode-electrolyte interface than with fluid flow. While the measured steady-state current of $\sim 0.95 \mu\text{A}$ agrees closely with the theoretical prediction of $0.914 \mu\text{A}$, this comparison alone isn't insufficient to confirm electroosmotic flow given the decaying trace shape. This result is therefore deemed inconclusive.

As a supplementary verification method, a vacuum pump was applied to one reservoir while the opposing reservoir contained fluid. Visual observation of fluid movement through the channel provided a qualitative confirmation of channel continuity independent of the electrical measurements.

A second round of current monitoring was conducted after leaving all samples submerged in DI water over the weekend. Testing focused on the bottom channel of glass Sample 6. Upon visual inspection, the channel interior was not distinctly visible, suggesting the microchannel was already fluid-filled, which is a qualitative indicator of channel continuity. DI water was evacuated from both reservoirs and replaced with 10 mM KCl as before.

A 500 V measurement was conducted first over a two-minute window. As shown in Figure 34, the current rose from approximately 0.3 nA and stabilized near 0.6 nA. The initial transient followed by stabilization may reflect gradual equilibration of the ionic concentration profile within the channel. Regardless, the sustained nanoampere-range signal falls well below the microampere levels associated with confirmed electroosmotic flow in the SiO₂ channel.

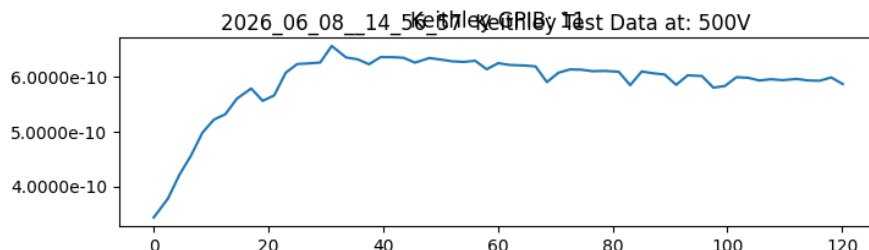
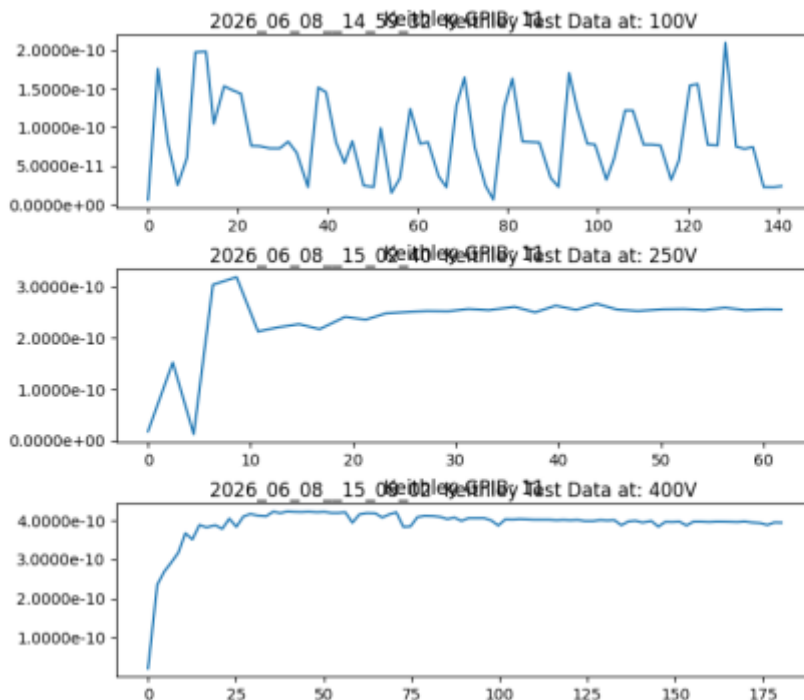


Figure 34: Current monitoring trace of glass Sample 6 (bottom channel) at 500 V, 10 mM KCl, showing stabilization near 0.6 nA over a two-minute window.

A subsequent measurement at 100 V exhibited only a transient current spike in the pA to nA range, with no stable plateau observed. This behavior suggests that the measured signal was dominated by noise, and means that no fluid was flowing. The applied voltage was then increased to 250 V, resulting in a stable current of approximately 0.3 nA. Increasing the voltage further to 400 V over a one-minute acquisition period produced a stable current of approximately 0.45 nA. The ratio of these currents ($0.45 \text{ nA} / 0.3 \text{ nA} \approx 1.5$) closely matches the ratio of the applied voltages ($400 \text{ V} / 250 \text{ V} = 1.6$), which is consistent with the expected linear relationship between current and applied electric field for a fixed channel geometry and electrolyte concentration. Representative current traces for all tested voltage conditions are shown in Figure 35.



With 20 mM KCl loaded into the channel, the applied voltage was reduced to 200 V and then increased to 250 V. Under both conditions, only transient current spikes in the picoampere-to-nanoampere range were observed, with no stable current plateau developing. This behavior closely mirrored the response previously observed at 100 V, suggesting the existence of a voltage threshold below which sustained transport is not achieved.

When considered alongside the apparent insensitivity of the measured current to electrolyte concentration, these results suggest that the nanoampere-level signals observed at higher voltages are unlikely to represent genuine electroosmotic flow. One possible explanation is that channel defects, partial blockages, or fabrication-related imperfections significantly restrict fluid transport, resulting in flow rates and corresponding currents that are orders of magnitude lower than expected. However, confirming this hypothesis would require additional characterization. Future work could include fluorescence-based flow visualization using microscopy to directly observe fluid motion within the microchannel and verify whether electroosmotic flow is occurring.

Across both rounds of testing, only one sample, a SiO₂ microchannel, produced microampere-level currents that were nominally consistent in magnitude with theoretical predictions. However, the shape of the current trace differed from the stable plateau expected during steady-state electroosmotic flow, making the result inconclusive. Additional testing on the bottom channel of glass Sample 6 showed stable nanoampere-level currents that scaled approximately linearly with applied voltage, but the measured values remained three to four orders of magnitude below theoretical predictions. Furthermore, increasing the KCl concentration from 10 mM to 20 mM produced little to no increase in current, contrary to expectations based on bulk electrolyte conductivity. These observations suggest that the measured currents were not dominated by fluid flow through the microchannel. All other fabricated channels across the SiO₂, glass, and PDMS devices similarly produced only picoampere to nanoampere level signals. The consistent low-current responses observed across these samples likely reflect errors in microchannel fabrication that cause issues such as leakage currents, channel blockage, or incomplete channel sealing rather than electroosmotic flow. Taken together, these results indicate that electroosmotic flow was not conclusively demonstrated in any fabricated device. Future work should focus on improving channel fabrication and bonding reliability and characterizing channel integrity.

7. Conclusion

The objective of this laboratory sequence was to fabricate, bond, and test three different types of microfluidic devices: SiO₂ microchannels, glass microchannels, and PDMS microchannels. Overall, the fabrication process was successful, and the majority of the targeted dimensions were achieved. Thermal oxidation, photolithography, BHF etching, and KOH etching generally produced channel depths and widths close to their design values. Although some discrepancies were observed, these were typically attributable to process variations, calibration uncertainties, or differences in wafer material properties. Throughout fabrication, reflectometry, profilometry, and microscopy proved valuable for verifying process success and understanding the effects of individual fabrication parameters on final device geometry.

Bonding was performed for all three device types, and microchannels were visually confirmed through microscopy and profilometry measurements. However, bonding also appeared to be one of the largest contributors to reduced device yield. Potential issues included PDMS deformation into channels, trapped air gaps, dust contamination, and defects introduced during reservoir formation with the biopsy punch. Because microfluidic device performance is highly dependent on channel integrity, even small bonding defects can significantly impact experimental results.

Current-monitoring experiments produced mixed results. Most devices exhibited only pA to nA level currents, suggesting little to no electroosmotic flow. One SiO₂ device produced microamp-level currents when higher voltages were applied, and currents remained measurable when the voltage was later reduced. However, the observed current response did not fully match the expected current-monitoring behavior and instead resembled a decaying signal. One possible explanation is that leakage currents through the silicon substrate or defects in the oxide layer contributed to the measured current. Increasing oxide thickness or modifying the oxidation process may help reduce this effect in future devices.

One glass microchannel produced current-monitoring trends that more closely resembled the expected electroosmotic flow response. Current increased with applied voltage as predicted by theory, although the measured dependence on electrolyte concentration was weaker than expected. Additionally, visual observations suggested that fluid transport may have occurred, as channels that were visible before filling were no longer easily distinguishable after being left overnight with electrolyte solution. Fluid movement was also observed qualitatively during preliminary testing when pressure was applied across channel reservoirs.

This laboratory sequence provided valuable experience in cleanroom fabrication, metrology, bonding, and device testing. The results demonstrated that successful microfluidic devices can be fabricated using the processes studied in this course, while also highlighting how sensitive device performance is to fabrication quality and process control. Future work should focus on improving process consistency by using a single wafer type throughout fabrication, incorporating reservoir features directly into future mask designs, improving bonding reliability, minimizing contamination, and incorporating deeper channel geometries. Increasing channel depth may improve fluid

transport and make electroosmotic flow easier to detect through current-monitoring measurements. These improvements would likely increase device yield, improve data quality, and result in more reliable characterization of microfluidic flow behavior.

Appendix A: Silicon Oxide Microchannel Fabrication Procedure

This appendix provides a complete step-by-step fabrication procedure for SiO₂ microfluidic channel devices. These steps are included for reproducibility and process documentation purposes.

A.1 Substrate Preparation and Sample Collection

Silicon substrates were prepared by cleaving a 3-inch silicon wafer into individual samples using a diamond scribe and wafer-cleaving pliers. Scratches were made on the polished side of the wafer, and the wafer was fractured along the scored lines. Resulting samples ranged from approximately 1.5 in × 1.5 in to 2 in × 2 in. Additional samples were provided by the lab manager. All samples were labeled on the unpolished side and inspected for defects before storage in sample carriers.

A.2 Solvent Cleaning

All samples were cleaned using the following sequence:

- Acetone ultrasonic bath for approximately 2 minutes
- Isopropyl alcohol (IPA) ultrasonic bath for approximately 2 minutes
- DI water rinse for 30–60 seconds
- Nitrogen gun drying

A.3 RCA SC-1 Clean

The SC-1 solution consisted of:

$$234 \text{ mL DI water} + 30 \text{ mL NH}_4\text{OH} + 36 \text{ mL H}_2\text{O}_2$$

The solution was heated to 60–65°C. Samples were immersed for 5 minutes, followed by a 2-minute DI rinse and nitrogen drying.

A.4 Buffered HF (BHF) Native Oxide Removal

A 10-second dip in buffered HF was used to remove native oxide prior to thermal oxidation. Samples were rinsed in DI water for 3 minutes and dried immediately afterward.

A.5 Thermal Oxidation

Wet oxidation was performed at 1030°C to grow approximately 6000 Å of SiO₂. To grow 6000 Å, the substrates must be placed in the oxidation furnace for about 1 hour 30 minutes and 4 seconds. Oxide thickness was verified using a reflectometer (Filmetrics F20) at multiple locations on each sample.

A.6 Photolithography Process

All samples underwent a standard photolithography sequence to define microchannel patterns prior to wet etching. The process was performed as follows:

1. **Dehydration Bake:** Samples were baked at 120°C for 3 minutes to remove surface moisture and improve resist adhesion.
2. **HMDS Treatment:** Hexamethyldisilazane (HMDS) vapor deposition was performed in a sealed chamber for 3 minutes to promote photoresist adhesion to the substrate surface.
3. **Photoresist Spin Coating:** AZ P4110 positive photoresist was spin-coated at 4000 RPM for 30 seconds, resulting in a film thickness of approximately 1.0–1.3 μm .
4. **Soft Bake:** Samples were soft-baked at 95°C for 1 minute to evaporate residual solvent.
5. **UV Exposure:** Performed using a Karl Süss MJB3 mask aligner with a three-channel chrome-on-glass mask featuring a channel width of 300 μm and a channel length of 2 cm. Before exposure, conformal mask contact was verified through optical inspection by confirming the absence of shadow features along the channel edges. Exposure dose was determined using the relationship: $\text{dose} = \text{irradiance} \times \text{time}$. With a target exposure dose of 90 mJ/cm^2 and a calibrated irradiance of 4.5 mW/cm^2 , the required exposure time was calculated as: $t = 90 \text{ mJ}/\text{cm}^2 \div 4.5 \text{ mW}/\text{cm}^2 = 20 \text{ s}$.
6. **Development:** Samples were developed in AZ 400K: DI (1:4) for 70 seconds with mild agitation, followed by a DI water rinse and nitrogen drying.
7. **Hard Bake:** A hard bake at 120°C for 3 minutes was performed to improve photoresist strength prior to etching.
8. **O₂ Plasma Descum:** Oxygen plasma treatment was used to remove residual photoresist and organic contaminants from exposed regions, ensuring clean feature definition before wet etching.

A.7 BHF Etching of SiO₂

Buffered HF was used to etch exposed SiO₂. Etch time was determined from calibration experiments correlating oxide thickness removal rate with time. This resulted in a time of 5 minutes to approximately remove all SiO₂. Samples were visually inspected to ensure that all SiO₂ was removed. However, our visual inspection went poorly, as in the next step our KOH etching was affected.

A.8 KOH Silicon Etching

Before KOH etching, residual photoresist was removed from the sample surface through a solvent clean. Anisotropic silicon etching was then performed in a KOH bath maintained at approximately 80°C. The etching solution was prepared using approximately 200 mL of 45% KOH solution mixed with approximately 100 mL of DI water. The exact concentration is uncertain because the volume specified in the Teaching Cleanroom SOP exceeded the capacity of the available beaker. As a result, preparation of the solution was stopped before the full volume could be added, and the concentration was adjusted accordingly.

An additional challenge was encountered during temperature stabilization of the KOH bath. The solution required substantially longer than anticipated to reach the target temperature of 80°C, preventing the etch from being performed on the originally scheduled day. The solution was therefore left in the beaker overnight and reheated the following day until the target temperature was reached. Once thermal equilibrium was established, the KOH etching process was carried out

according to the planned procedure.

Following initial etch attempts, profilometry measurements indicated no measurable change in channel depth, leading to the conclusion that a residual SiO_2 layer remained at the channel floor, preventing KOH from reaching the underlying silicon. To address this, samples were dipped in BHF for an additional minute to ensure complete oxide removal. Given that BHF etches isotropically and the majority of the SiO_2 had already been removed in the preceding step, it was determined that continued etching until bare silicon was confirmed would not adversely affect the channel geometry. This decision was further supported by considering the estimated remaining SiO_2 thickness in relation to the known selectivity of KOH for Si over SiO_2 , which confirmed that any residual oxide exposure would be negligible.

Once the silicon substrate was confirmed, the KOH etch rate was determined experimentally through profilometry-based calibration. The etch duration required to achieve the 4 μm target channel depth was determined to be approximately 4 minutes and 22 seconds. A single calibration sample was etched first and verified by profilometry before the remaining samples were processed under the same conditions.

A.9 Hard Mask Removal

Samples were first subjected to a solvent clean, after which the remaining SiO_2 was removed via a 5-minute 30-second BHF dip. Complete oxide removal was verified through visual inspection. The dip duration intentionally exceeded the time calculated from calibration to ensure full removal of any residual oxide. This over-etch is acceptable given the high selectivity of BHF for SiO_2 over Si, meaning that extended exposure does not risk damage to the underlying silicon substrate.

A.10 Final Thermal Oxidation

A final thermal oxidation step was performed to grow approximately 6,500 Å of SiO_2 within the microchannels. The oxidation sequence consisted of an initial dry oxidation for 10 minutes, followed by wet oxidation for 1 hour 42 minutes and 49 seconds, and a final dry oxidation step for 10 minutes. Oxide thickness was subsequently verified using reflectometry and profilometry.

Due to the comparatively low growth rate of dry oxidation, the oxide thickness contributed by the dry oxidation steps is negligible relative to that grown during the wet oxidation phase, and was therefore not included when evaluating the total oxide thickness against the 6,500 Å target. Despite contributing minimally to overall thickness, the dry oxidation steps serve an important functional role. This is because dry oxidation produces a denser, more uniform oxide compared to wet oxidation, making it the preferred method for defining the final channel surface that will be in direct contact with fluid during device operation.

A.11 Bonding SiO_2 channels to PDMS

The first step in bonding the SiO_2 microchannels was the preparation of a flat PDMS layer. PDMS was

prepared by mixing 10 mL of Part A with 1 mL of Part B in a plastic weighing boat. The mixture was stirred for approximately 10 minutes to ensure a homogeneous solution and promote uniform curing.

Before PDMS casting, a flat silicon substrate was solvent cleaned. The mixed PDMS was then poured onto the substrate, ensuring uniform coverage across the entire surface. The substrate was subsequently placed in a vacuum chamber for approximately 1 hour to remove trapped air bubbles. Following degassing, the substrate was transferred to a hot plate and cured at 80°C for 2 hours. Care was taken to keep the substrate level during curing to ensure a uniform PDMS thickness across the sample.

After curing, the PDMS was cut into rectangular pieces approximately 2.5 cm × 2.5 cm in size to fully cover the microchannels. The PDMS was then carefully peeled from the silicon substrate using tweezers and an X-Acto knife.

Irreversible bonding was performed using O₂ plasma activation. Immediately after removal from the mold, both the PDMS and the SiO₂ microchannel substrate were placed in the O₂ plasma descum system for 2 minutes. Following plasma treatment, the PDMS was carefully aligned and placed over the microchannels on the SiO₂ substrate. Gentle pressure was applied using tweezers to remove trapped air bubbles and promote conformal contact between the surfaces while avoiding direct pressure over the channels.

After bonding, reservoirs were created using a biopsy punch. The punch was aligned near the ends of each microchannel and pressed downward until contact was made with the silicon substrate. A slight twisting motion was then applied to facilitate clean removal of the PDMS plug. In cases where the punched PDMS did not fully separate, tweezers were used to carefully remove the remaining material. This process was repeated for all channel inlet and outlet reservoirs.

Once all reservoirs had been formed, the bonded device was placed on a 120°C hot plate for 5 minutes to strengthen adhesion between the PDMS and SiO₂ surfaces. After cooling to room temperature, the device was considered ready for current-monitoring experiments.

Appendix B: Glass Microchannel Fabrication

This appendix provides a complete step-by-step fabrication procedure for glass microfluidic channel devices. These steps are included for reproducibility and process documentation purposes.

Glass microfluidic channels were fabricated using the same three-channel mask design as the silicon process (300 μm wide, 2 cm long channels). Unlike silicon, glass was etched using buffered HF (BHF), resulting in isotropic etching and rounded channel profiles due to lateral undercut beneath the photoresist mask. The target channel depth for the glass devices was 2 μm.

B.1 Substrate Preparation and Cleaning

Standard microscope glass slides were cleaved into approximately 2 in × 2 in samples using a

diamond scribe and cleaving pliers. Six samples were prepared. Each sample was labeled on the non-process side and inspected for defects before processing.

All glass substrates underwent solvent cleaning and RCA SC-1 cleaning following the same procedures used in the silicon fabrication process.

B.2 Photolithography

Photolithography was performed using the same process flow, materials, and parameters described in the silicon fabrication section (dehydration bake, HMDS treatment, AZ P4110 spin coating, soft bake, UV exposure, development, hard bake, and O₂ plasma descum).

A key observed difference was during spin coating: samples processed using a larger chuck exhibited photoresist wrap-around on the backside of the wafer, leading to non-uniform edge coverage. Use of a smaller chuck significantly improved uniformity and cleanliness of the wafer.

B.3 BHF Etching of Glass Channels

Buffered HF (BHF) was used to etch exposed regions of the glass substrate. Due to the isotropic nature of HF-based glass etching, material removal occurred both vertically and laterally, producing rounded, U-shaped channel profiles and measurable undercut beneath the resist mask.

A dedicated calibration sample was used to determine the glass etch rate. The sample was etched in multiple timed intervals, with profilometry measurements taken after each step. Channel depth was plotted versus cumulative etch time to determine the effective etch rate.

Using the calibrated etch rate, all remaining samples were etched in BHF for approximately 4 minutes and 40 seconds to achieve the target 2 μm channel depth. Samples were rinsed in DI water for 3 minutes and dried using nitrogen gas after rinsing.

B.4 Bonding glass channels to PDMS

The same bonding procedure used for the SiO₂ microchannels was employed for the glass microchannels and is described in *Appendix A.4*. During reservoir formation, one biopsy punch operation resulted in tearing of the PDMS near a reservoir, rendering one of the channels unusable. This defect may also create leakage pathways that could negatively affect fluid transport and current-monitoring measurements.

Despite this issue, visual inspection suggested that two channels remained intact and were potential candidates for successful fluid flow. No major bonding defects were observed in these channels before testing, indicating that they could still be viable for current-monitoring experiments.

Appendix C PDMS Microchannel Fabrication Procedure:

The fabrication procedure for the PDMS microchannel devices was largely identical to the SiO₂ microchannel fabrication process described in *Appendix A*. This appendix documents only the steps and parameters that differed from that procedure.

C.1 Substrate Preparation and Sample Collection

Substrate preparation followed the procedure described in *Appendix A.1*. Six samples were cleaved to approximately 1.75 in × 1.75 in.

C.2 Solvent Cleaning

Solvent cleaning was performed as described in *Appendix A.2*.

C.3 RCA SC-1 Clean

RCA cleaning was performed as described in *Appendix A.3*. During this step, Sample 4 broke in half and was excluded from all subsequent processing.

C.4 Buffered HF (BHF) Native Oxide Removal

BHF native oxide removal was performed as described in *Appendix A.4*.

C.5 Thermal Oxidation

Thermal oxidation was performed as described in *Appendix A.5*, with the exception that the target oxide thickness was approximately 3,500 Å rather than 6,000 Å. To achieve this, the oxidation sequence consisted of an initial dry oxidation for 10 minutes, followed by wet oxidation for 38 minutes and 23 seconds, and a final dry oxidation step for 10 minutes. Samples 5 and 6 were double-sided and potentially doped, and exhibited slightly greater oxide thicknesses than the single-sided samples due to oxidation occurring on both surfaces.

C.6 Photolithography

Photolithography was performed as described in *Appendix A.6*, with the exception that a different mask was used corresponding to the PDMS mold channel geometry. Post-exposure profilometry was performed to verify resist thickness before proceeding with the remaining samples.

C.7 BHF Etching of SiO₂ Hard Mask

Samples were immersed in BHF for approximately 5 minutes and 30 seconds. Progress was monitored by visual inspection, and etching was continued until the SiO₂ was confirmed to be fully removed from all exposed regions. Care was taken to avoid over-etching, as prolonged BHF exposure risks undercutting the photoresist and compromising feature geometry. Once oxide removal was complete, the photoresist was stripped and the samples were prepared for KOH etching.

C.8 KOH Silicon Etching

KOH etching was performed as described in *Appendix A.8*, with the following differences. The target mold depth was 3 μm rather than 4 μm . Sample 6 was selected as the calibration sample and etched for 4 minutes to establish the etch rate. Based on this calibration, Samples 1–4 were etched for approximately 4 minutes and 10 seconds, and Sample 5 was etched for approximately 4 minutes. Final mold depths were verified by profilometry and were generally close to the 3 μm target. Detailed analysis is presented in the PDMS Mold KOH Etching section of the Process Results.

C.9 Hard Mask Removal

Hard mask removal was performed as described in *Appendix A.9*. All samples were immersed in BHF for approximately 5 minutes to remove the remaining sacrificial SiO_2 . After the oxide strip was completed, the samples were rinsed, dried, and stored until PDMS casting.

C.10 PDMS Mold Casting

Mold substrates were solvent cleaned prior to casting. PDMS was prepared and cast as described in *Appendix A.11*, with the silicon mold substrate placed on aluminum foil with edges curled upward to form a containment vessel. The mixed PDMS was poured to a thickness of less than 5 mm. The cured PDMS was left on the mold until immediately before bonding to prevent dust contamination.

C.11 Reservoir Formation and Bonding to Glass Substrates

Bonding was performed as described in *Appendix A.11*, with one key difference: reservoir holes were created using a biopsy punch prior to O_2 plasma treatment rather than after bonding. This was possible because the PDMS mold design incorporated visible reservoir features, allowing punch locations to be identified before bonding.

Microscopy after bonding indicated good structural integrity with no major defects or obstructions for most channels. As a qualitative verification, a water droplet introduced into one channel produced a visible meniscus extending through the channel, confirming the channel remained open and capable of fluid transport. An image demonstrating this behavior is provided in the Bonding section.

Following verification, the devices were stored until current-monitoring experiments were performed.

Appendix D: Mask Designs

All mask designs used throughout the lab can be seen below in Figure 37 and 38. Figure 37 represents mask used to fabricate the SiO_2 and glass microchannel devices. Figure 38 represents the mask used to fabricate the PDMS mold used to create PDMS microchannel devices.

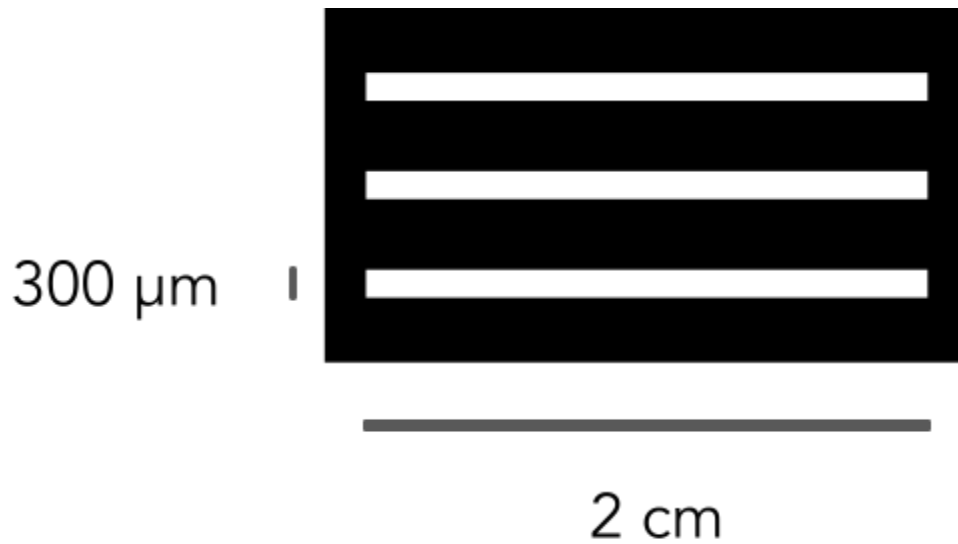


Figure 37: 3 Channel Mask Design(SiO_2 and Glass Microchannel Mask)

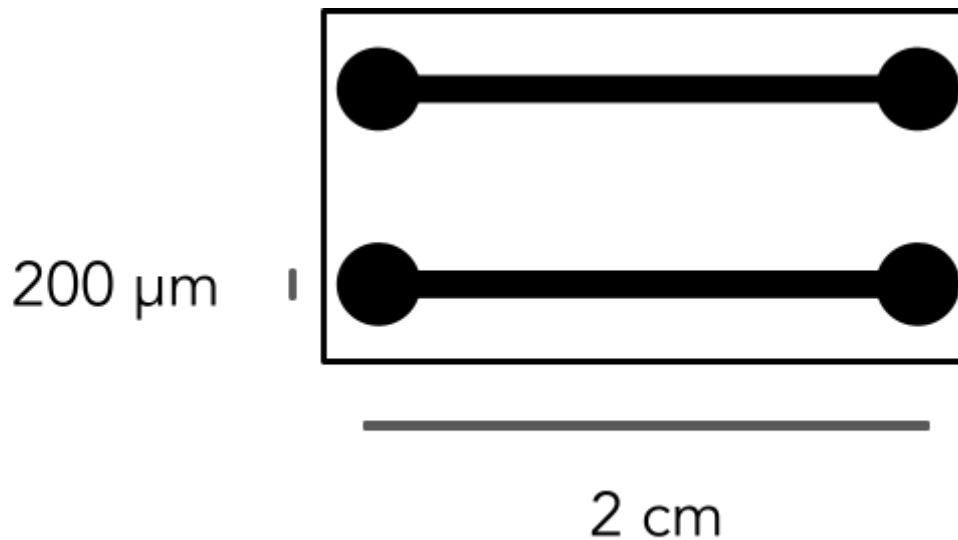


Figure 38: 2 Channel Mask Design(PDMS mold Mask)

Appendix E: Approximate Profilometer Traces

Approximate profilometer traces used to acquire measurements such as width, sidewall angle, and depth used traces seen below in Figure 39 and 40 for their respective chip.

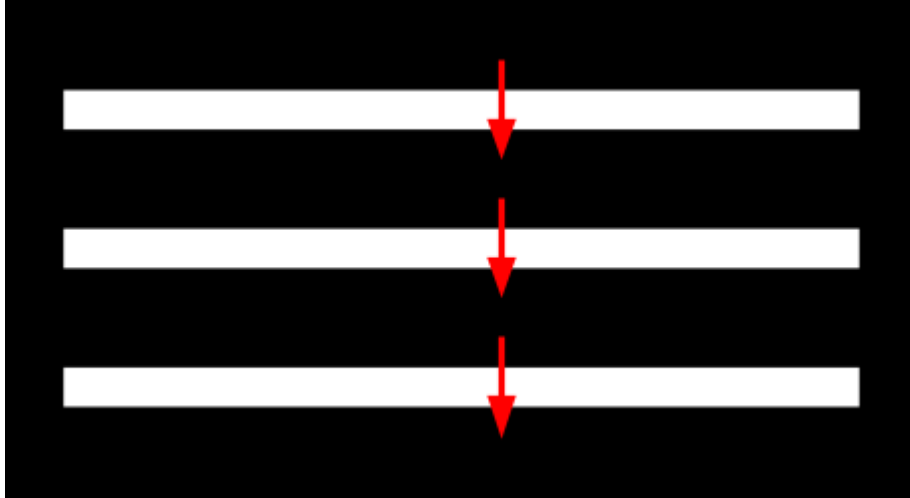


Figure 39: 3-channel chip design profilometer traces:

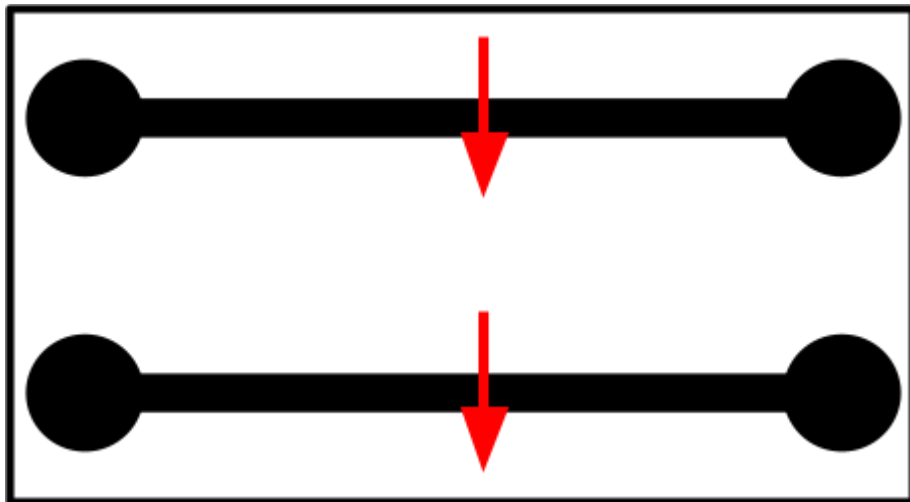


Figure 40: 2-channel chip design profilometer traces: